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# RAGGER: One-shot reasoning with Hyperlayerd Asynchronous Hybrid RAG

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## Abstract

While Retrieval-Augmented Generation (RAG) has demonstrated significant potential in enhancing Large Language Models through external knowledge integration, current implementations face substantial trade-offs between efficiency and performance, particularly when scaling to large documents with millions of tokens. We present Hyper-layered Asynchronous Hybrid RAG (HAH-RAG), a novel framework that implements parallel hybrid search with asynchronous vector database querying. Our approach executes sparse and dense retrievals across multiple specialized layers, each optimized for different granularities of information retrieval. Through asynchronous vector database calls, we achieve a significant latency reduction while maintaining comprehensive search capabilities. Empirical evaluation demonstrates that HAH-RAG achieves a 66.6% reduction in energy consumption and CO<sub>2</sub> emissions compared to traditional hybrid approaches, while maintaining superior performance in factuality and correctness, scoring 0.745 and 0.645, respectively. The method demonstrates particular effectiveness in response time, achieving an average latency of 4.47 seconds - an improvement of 64.5% over existing methods. These results establish HAH-RAG as a promising direction for frugal implementation of RAG without compromising the quality of the performance.

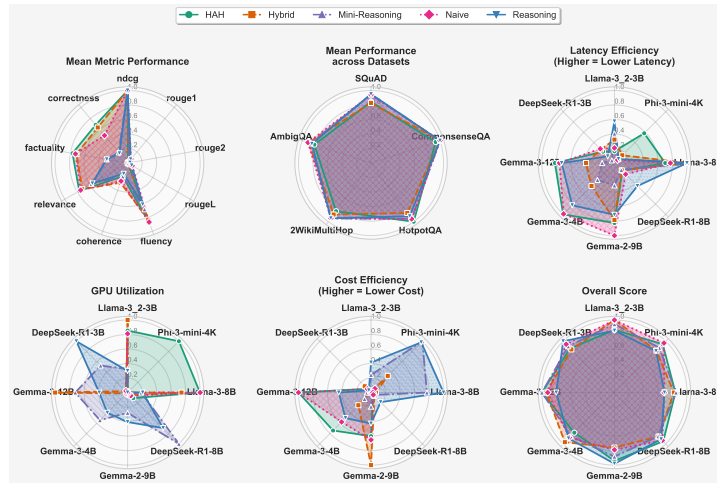


Figure 1: Performance comparison of pipeline strategies across quality, efficiency, and resource metrics. Radar plots summarize mean metric performance (e.g., NDCG, ROUGE, correctness, factuality, relevance, coherence, and fluency), average performance across QA datasets, latency efficiency, GPU utilization, cost efficiency, and overall score for each pipeline. HAH consistently demonstrates strong and well-balanced performance, achieving favorable trade-offs between answer quality, latency, and computational efficiency across models.

## 1 Introduction

Retrieval-Augmented Generation (RAG) has emerged as one of the most significant advances in AI, offering up-to-date and reliable external knowledge to enhance various tasks [Lewis et al. \(2020\)](#). In the era of AI-Generated Content (AIGC), RAG’s powerful retrieval advantages enable existing generative AI models to produce higher-quality output by incorporating an additional knowledge base [Li et al. \(2023\)](#); [Sheynin et al. \(2023\)](#); [Wu et al. \(2024\)](#). Large Language Models (LLMs) have demonstrated exceptional abilities in language understanding and generation [Achiam et al. \(2023\)](#); [Chowdhery et al. \(2023\)](#), paving the way for their adoption in various decision-making industries such as the semiconductor ([Ezukwoke et al., 2024](#)). However, they still face inherent limitations, including hallucinations and outdated internal knowledge [Zhao et al. \(2023, 2024\)](#). These challenges are particularly prominent in domain-specific fields like medicine and law, where recent studies have shown hallucination rates ranging from 69% to 88% in responses to specific legal queries [Dahl et al. \(2024\)](#).

To address these limitations, RAG has been integrated with LLM to enhance their performance in various tasks [Borgeaud et al. \(2022\)](#); [Izacard and Grave \(2021a\)](#); [Khandelwal et al. \(2020\)](#); [Shi et al. \(2023\)](#). RAG-augmented LLMs have shown particular promise in tasks that demand current and reliable knowledge, such as Question Answering, AI4Science, and software engineering. For example, [Lozano et al. \(2023\)](#) introduced a scientific QA method based on dynamic scientific literature retrieval, while MolReGPT leverages RAG to enhance ChatGPT’s in-context learning ability for molecular discovery [Li et al. \(2023\)](#). Integration of RAG with LLM has also been shown to be effective in reducing hallucinations in conversational tasks [Shuster et al. \(2021\)](#); [Xu et al. \(2022\)](#). As illustrated in the research, RAG-augmented dialogue methods can provide more accurate answers by retrieving relevant knowledge from external databases when dealing with out-of-scope queries.

**Contributions.** This work makes the following contributions (see Figure 1 for a visual summary):

1. We propose HAH-RAG, a novel retrieval-augmented generation framework implementing asynchronous multi-layer hybrid search with dynamic parameter adaptation and contextual compression.
2. We provide comprehensive empirical evaluation across five benchmarks and eight language models, demonstrating consistent improvements in factuality and relevance while reducing computational overhead.
3. We analyze the interplay between retrieval architecture, model characteristics, and downstream generation quality, identifying conditions under which sophisticated retrieval strategies yield maximum benefit.
4. We quantify the environmental impact of pipeline design choices, demonstrating that HAH-RAG reduces energy consumption by approximately 66% compared to traditional hybrid approaches.

## 2 Background

Retrieval-Augmented Generation (RAG) has emerged as a key advancement in artificial intelligence, offering a solution to enhance Large Language Models (LLMs) by incorporating up-to-date and reliable external knowledge [Lewis et al. \(2020\)](#); [Borgeaud et al. \(2022\)](#). Although LLMs have demonstrated remarkable abilities in language understanding and generation, they face inherent limitations, including hallucinations, outdated knowledge, and non-transparent reasoning processes [Zhao et al. \(2023, 2024\)](#).

The state-of-the-art RAG-LLMs have shown remarkable success across various domains. In molecular discovery, RetMol [Wang et al. \(2023b\)](#) and MoleculeSTM [Liu et al. \(2023\)](#) leverage RAG for improved molecular property prediction and generation. Recent works by [Li et al. \(2024\)](#); [Yepes et al. \(2024\)](#); [Zhang et al. \(2023\)](#) have also demonstrated the effectiveness of RAG-LLMs in financial sentiment analysis and document processing. In the software engineering domain, advanced methods such as Atlas [Nashid et al. \(2023\)](#) and DocPrompting [Zhou et al. \(2022\)](#) have shown significant improvements in code generation and program repair tasks.

RAG addresses these challenges by enabling LLMs to dynamically access and integrate knowledge from external sources during generation.

Table 1: Evolution of RAG methods, Retriever, Granularity and Integration with the pros and cons for each.

| Category    | Type/Paradigm      | Key Features                                                                        | Considerations                                                                                                                                                            |
|-------------|--------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evolution   | Naive RAG          | - Basic indexing<br>- Simple retrieval<br>- Direct generation                       | <b>Pros:</b> Simple implementation, low overhead<br><b>Cons:</b> Limited accuracy, prone to hallucinations<br><b>Use-case:</b> Simple QA, basic retrieval                 |
|             | Advanced RAG       | - Enhanced indexing<br>- Pre/post retrieval processing<br>- Optimized generation    | <b>Pros:</b> Better accuracy, improved context handling<br><b>Cons:</b> Higher computational cost<br><b>Use-case:</b> Knowledge-intensive tasks, reasoning                |
|             | Modular RAG        | - Specialized modules<br>- Flexible architecture<br>- Advanced integration patterns | <b>Pros:</b> High adaptability, task-specific optimization<br><b>Cons:</b> method complexity, resource intensive<br><b>Use-case:</b> Complex reasoning, multi-modal tasks |
| Retriever   | Sparse Retrieval   | - Word-based (TF-IDF, BM25)<br>- No training required<br>- Direct term matching     | <b>Pros:</b> Simple, interpretable results<br><b>Cons:</b> Limited semantic understanding                                                                                 |
|             | Dense Retrieval    | - Embedding-based<br>- Learned representations<br>- Semantic matching               | <b>Pros:</b> Better semantic understanding<br><b>Cons:</b> Training required, computationally intensive                                                                   |
| Granularity | Document-level     | - Full document retrieval<br>- Comprehensive context                                | <b>Pros:</b> Complete context coverage<br><b>Cons:</b> Information overload                                                                                               |
|             | Chunk-level        | - Segmented retrieval<br>- Balanced context size                                    | <b>Pros:</b> Good efficiency balance<br><b>Cons:</b> Potential fragmentation                                                                                              |
|             | Entity-level       | - Knowledge graph based<br>- Structured information                                 | <b>Pros:</b> High precision<br><b>Cons:</b> Limited to structured data                                                                                                    |
| Integration | Input-Layer        | - Content concatenation<br>- Direct augmentation                                    | <b>Pros:</b> Easy implementation<br><b>Cons:</b> Input length limitations                                                                                                 |
|             | Output-Layer       | - Post-processing<br>- Result combination                                           | <b>Pros:</b> Flexible implementation<br><b>Cons:</b> Limited reasoning depth                                                                                              |
|             | Intermediate-Layer | - Deep integration<br>- Advanced processing                                         | <b>Pros:</b> Better reasoning<br><b>Cons:</b> Complex implementation                                                                                                      |

The table 1 provides a comprehensive overview of the key components and considerations in RAG methods. The evolution from Naive RAG to Modular RAG shows the increasing sophistication of these methods, while the various retrieval methods and integration strategies offer different approaches to optimize performance based on specific use cases and requirements.

## 2.1 Evolution of RAG Methods

The development of RAG can be categorized into three distinct paradigms, namely: Native RAG, Advanced RAG, and Modular RAG.

### 2.1.1 Naive RAG

The initial RAG paradigm emerged as a straightforward "Retrieve-Read" framework [Ram et al. \(2023\)](#), implementing a three-stage process: indexing, retrieval, and generation. During indexing, documents are processed into a uniform format, segmented into manageable chunks, and converted to vector representations for efficient similarity searches [Gao et al. \(2023b\)](#). The retrieval phase employs the same encoding model to transform queries into vectors, computing similarity scores with indexed chunks. Finally, the generation stage synthesizes the query and retrieved documents into a coherent prompt for the language model response [Lewis et al. \(2020\)](#).

However, Naive RAG faces several limitations: retrieval precision and recall challenges often result in misaligned or irrelevant chunks, while the generation phase can suffer from hallucination issues where content is produced without proper support from the retrieved context [Asai et al. \(2023\)](#). In addition, the integration of the information retrieved with different tasks can be challenging, sometimes resulting in disjointed or incoherent output.

### 2.1.2 Advanced RAG

Advanced RAG emerged to address the limitations of its predecessor by introducing sophisticated optimization strategies around pre-retrieval and post-retrieval processes [Izacard and Grave \(2021b\)](#). Key improvements include improved indexing techniques through sliding windows, fine-grained

segmentation, and metadata incorporation [Gao et al. \(2023b\)](#). Pre-retrieval optimization focuses on improving index structures and query transformation, while post-retrieval processes include reranking and context compression to improve the quality of retrieved information [Ma et al. \(2023\)](#).

This paradigm introduces query optimization techniques such as query rewriting, transformation, and expansion [Wang et al. \(2023a\)](#); [Gao et al. \(2023a\)](#). Post-retrieval enhancements include sophisticated reranking strategies and context compression methods to improve the relevance and usability of retrieved information [Hofstätter et al. \(2023\)](#).

### 2.1.3 Modular RAG

The most advanced paradigm, modular RAG, offers enhanced adaptability and versatility by incorporating various strategies to improve its components [Asai et al. \(2023\)](#). This approach introduces specialized modules for similarity searches, retriever fine-tuning, and various other functionalities [Gao et al. \(2023b\)](#). Key innovations include: **Search Module**, which enables direct searches across various data sources using LLM-generated code and query languages. **Memory Module**, Leverages LLM’s memory to guide retrieval through iterative self-enhancement. **Task Adapter**, Tailors RAG to various downstream tasks through automated prompt retrieval and task-specific retriever creation [Cheng et al. \(2023\)](#) and **Predict Module** reduces redundancy and noise by generating context directly through LLM [Yu et al. \(2022\)](#).

## 2.2 Retrieval Methods

### 2.2.1 Retriever Types

Modern RAG methods employ three primary types of retrievers:

**Sparse Retrieval** : Implements word-based approaches using techniques such as TF-IDF and BM25 [Robertson et al. \(2009\)](#). These methods rely on inverted index matching with raw data input, offering straightforward implementation but limited by their fixed-term-based nature. Although effective for direct text matching, they struggle with semantic understanding and context awareness [Gao et al. \(2023b\)](#).

**Dense Retrieval** : Introduces neural networks to embed queries and documents in continuous vector spaces [Karpukhin et al. \(2020\)](#). Key implementations include

- **Bi-encoder Architecture**: Implements two separate BERT-based encoders for queries and documents, as demonstrated in DPR [Karpukhin et al. \(2020\)](#)
- **Single-encoder Models**: Pre-trained through contrastive learning on unaligned documents, offering better generalization (e.g., Contriever) [Gautier et al. \(2022\)](#)
- **Generation Model Integration**: Some methods use parts of the generation model as embedding layers to improve the alignment between the retrieval and generation processes [Lewis et al. \(2020\)](#)

**Hybrid Retrieval** Hybrid retrieval methods combine the strengths of sparse and dense retrieval approaches to enhance the overall performance of RAG [Ma et al. \(2023\)](#); [Gao et al. \(2023b\)](#). These methods leverage the complementary nature of lexical matching (sparse) and semantic understanding (dense) to improve retrieval accuracy and robustness. The sparse retriever implements traditional lexical matching using methods such as BM25 or TF-IDF [Robertson et al. \(2009\)](#), while the dense retriever uses neural encoders to capture semantic relationships [Karpukhin et al. \(2020\)](#).

Two main methods are used to combine the results of BM25 and dense retriever including **parallel** and **sequential retrievers**. Parallel processing is used to combine the results of both sparse and dense simultaneously in the input query, the results merged through a fusion mechanism [Khatab et al. \(2022\)](#); [Gao et al. \(2023a\)](#). Sequential processing allows the output of one retriever to guide or filter the other’s search space (See 2). Through adaptive switching, the method dynamically chooses between sparse and dense retrieval based on the query characteristics [Ram et al. \(2023\)](#). A Score-based Fusion method is used to combine the scores resulting from both methods as follows:

$$S_{final} = \alpha S_{sparse} + (1 - \alpha) S_{dense} \quad (1)$$

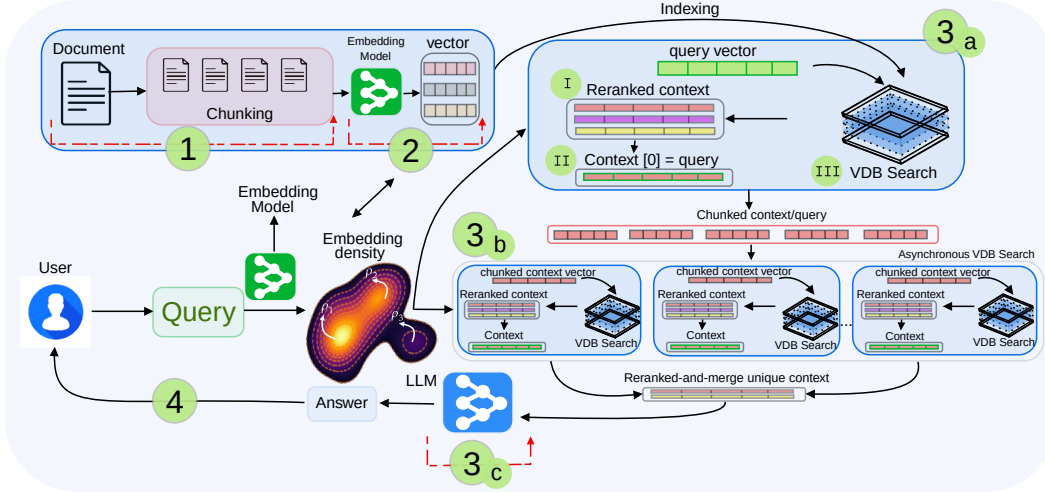


Figure 2: Proposed Hyper-layered Asynchronous Hybrid RAG (HAH-RAG).

Where,  $\alpha$  is a weight parameter that can be fixed based on empirical testing (trial-and-error), dynamically adjusted using query characteristics, or learned through supervised training. This fusion strategy gives the hybrid approach a significant edge in retrieval performance in both contextual understanding and robustness [Borgeaud et al. \(2022\)](#); [Lewis et al. \(2020\)](#).

### 2.3 Architecture Innovations

Recent architectural advances in RAG-LLMs have focused on three primary integration approaches:

1. **Input-Layer Integration:** Methods like In-Context RALM [Ram et al. \(2023\)](#) combine retrieved documents with the original input for joint processing. FID [Izacard and Grave \(2021b\)](#) improved on this by introducing independent document processing in the encoder, allowing scalable handling of multiple contexts.
2. **Output-Layer Integration:** methods like kNN-LM [Khandelwal et al. \(2020\)](#) implement post hoc integration by interpolating LM and retrieval-based distributions. REFEED [Yu et al. \(2023\)](#) enhanced this approach by introducing an answer refinement mechanism using LLMs to evaluate retrieved information.
3. **Intermediate-Layer Integration:** More sophisticated approaches like RETRO [Borgeaud et al. \(2022\)](#) introduced Chunked Cross Attention (CCA) layers for processing retrieved chunks within generator blocks, offering efficient handling of frequently retrieved information.

### 2.4 Training Strategies

State-of-the-art training approaches can be categorized into:

1. **Training-free Methods:** These approaches leverage retrieved knowledge during inference without additional training ([Izacard and Grave, 2021b](#); [Jiang et al., 2023](#); [Khattab et al., 2022](#)), making them computationally efficient and adaptable.
2. **Sequential Training:** Methods like RETRO [Borgeaud et al. \(2022\)](#) and RALMs, [Yoran et al. \(2023\)](#) adopts a staged approach, first training one component (retriever or generator) before optimizing the other, allowing better component specialization.
3. **Joint Training:** Advanced approaches like RAG, [Lewis et al. \(2020\)](#) and REALM, [Guu et al. \(2020\)](#) optimize both retriever and generator simultaneously, enabling better synergy between components.

Our proposed RAG scheme leverages the training of the retriever at the time of creating the index without the need of repeating inference.

### 3 Reasoning Template-based HAH

Traditional Retrieval-Augmented Generation approaches treat all queries uniformly, applying identical retrieval strategies regardless of the underlying cognitive demands of the question. This one-size-fits-all paradigm fundamentally limits system performance: a factual lookup query (“What is the capital of France?”) requires fundamentally different retrieval characteristics than a multi-hop reasoning question (“How did the economic policies of the 1920s contribute to the Great Depression?”). We introduce Reasoning Template-based Hyper-layered Asynchronous Hybrid RAG (HAH-RAG), a framework that addresses this limitation through principled reasoning-aware retrieval and generation. This may alternatively be described as an adaptive, template-guided HAH-RAG framework, operating in a similar manner to a chain-of-thought process (Wei et al., 2022).

#### 3.1 Motivation and Design Philosophy

The central insight motivating HAH-RAG is that human question-answering involves distinct cognitive processes depending on the nature of the inquiry. When faced with a factual question, humans retrieve specific facts from memory; analytical questions trigger systematic decomposition; comparative questions invoke parallel evaluation of alternatives (See Figure 2). Our framework operationalizes this observation by implementing a Bayesian reasoning classifier that dynamically identifies the cognitive demands of each query and adapts the retrieval-generation pipeline accordingly.

HAH-RAG achieves three fundamental objectives. First, it **optimizes the precision-speed trade-off** through reasoning-guided parallel retrieval, where retrieval parameters ( $k, \lambda$ ) are dynamically adjusted based on query complexity analysis. Second, it **enhances response quality** through context-aware reranking that incorporates both semantic coherence scores and reasoning-specific pattern analysis, ensuring that retrieved contexts align with the identified reasoning requirements. Third, it **maximizes computational efficiency** through asynchronous vector database search that parallelizes both sparse BM25 and dense retrieval operations, substantially reducing latency while maintaining comprehensive search coverage.

#### 3.2 Reasoning Type Taxonomy

The foundation of HAH-RAG’s adaptive behavior rests on a principled taxonomy of reasoning types. This classification emerges from the observation that different categories of questions impose fundamentally different demands on both retrieval and generation subsystems. A question asking for the capital of a country requires precise fact retrieval and minimal synthesis, whereas a question exploring causal relationships between historical events demands comprehensive evidence gathering and careful argumentative construction. By recognizing these distinctions at query time, HAH-RAG can tailor its entire processing pipeline to match the cognitive demands of each inquiry.

We identify five primary reasoning categories, each characterized by distinctive linguistic markers and associated with specific retrieval and generation strategies. **Factual reasoning** encompasses queries seeking specific, verifiable information. These queries typically begin with interrogative words such as **what**, **who**, **where**, or **when**, and expect precise, often singular answers. The retrieval strategy for factual queries prioritizes precision over recall, emphasizing documents with high lexical overlap to ensure accurate fact retrieval. **Analytical reasoning** involves systematic examination and interpretation of information, signaled by terms like **how**, **analyze**, **evaluate**, or **explain**. For these queries, the system emphasizes comprehensive context gathering and instructs the generation model to decompose problems into constituent parts before synthesizing a response.

**Comparative reasoning** requires evaluation of similarities and differences between entities or concepts. Detection relies on markers such as **compare**, **contrast**, **versus**, or **differ**, and triggers retrieval strategies that gather evidence about each comparison target independently before structured synthesis. **Causal reasoning** addresses cause-effect relationships, typically signaled by **why**, **because**, or **result**, and benefits from chain-of-thought retrieval where initial results inform subsequent queries to build complete causal chains. Finally, **hypothetical reasoning** handles conditional and speculative queries containing markers like **if**, **would**, or **assume**, prioritizing documents that discuss scenarios, outcomes, and conditional relationships.

Beyond these substantive categories, we implement a **trivial query bypass** mechanism that recognizes conversational inputs, greetings, acknowledgments, and farewells, through vocabulary match-

ing. Trivial queries bypass the expensive retrieval pipeline entirely, generating responses directly from the language model’s parametric knowledge. This optimization substantially reduces average latency for real-world workloads that intermix substantive queries with conversational exchanges.

The reasoning type taxonomy serves as the foundation for downstream adaptations throughout the HAH-RAG pipeline. Once a query’s reasoning type is identified (through the Bayesian classifier described in Section 4.1), the system adjusts retrieval parameters, selects appropriate fusion strategies, and instantiates reasoning-specific generation prompts. The complete specification of reasoning patterns and detection thresholds appears in Appendix C.1.

### 3.3 Two-Pass Retrieval Architecture

A critical limitation of conventional RAG systems is their reliance on single-pass retrieval, which often fails to capture the full context required for complex queries. Consider a multi-hop question such as *What economic policies contributed to the decline of the company that Steve Jobs founded?* A single retrieval pass might return documents about Apple’s history or about economic policies in isolation, but may miss the bridging documents that connect these topics. HAH-RAG addresses this limitation through a two-pass retrieval architecture that captures both primary evidence and supporting context.

The first pass performs initial context retrieval with parameters adapted to query complexity. The system analyzes each query through multiple heuristics, including word count, question count, and domain terminology presence, and determines retrieval parameters accordingly. For complex queries (those with more than 20 words or containing multiple questions), the system retrieves a larger initial set with  $k = 7$  documents and a diversity parameter of  $\lambda = 0.6$ . Simpler queries receive more focused retrieval with smaller values. For language models with extended context windows (those exceeding 8 billion parameters), these values are scaled by a factor of 10 to fully exploit the available context capacity.

The initial retrieval proceeds through three chained operations: ensemble retrieval combining sparse and dense methods, flash reranking to improve relevance ordering, and compression to eliminate redundancy. Formally, given a query  $q$  and retrieval count  $k$ , the first-pass contexts are obtained as:

$$C_1 = \text{Compress}(\text{Rerank}(\text{Ensemble}(q, k)), 0.7) \quad (2)$$

where the compression ratio of 0.7 retains the top 70% of reranked passages.

The second pass addresses a fundamental insight, initial retrieval results often contain implicit references to related concepts that, if pursued, would yield valuable additional context. To exploit this, HAH-RAG constructs a synthetic document by concatenating the first-pass contexts:

$$d_{\text{synth}} = c_1 \oplus c_2 \oplus \dots \oplus c_{|C_1|} \quad (3)$$

where  $\oplus$  denotes text concatenation. This synthetic document is segmented into chunks, each of which is submitted to parallel asynchronous search. The union of these second-pass results captures bridging documents that connect disparate pieces of evidence from the initial retrieval:

$$C_2 = \bigcup_{j=1}^n \text{Search}(\text{chunk}_j(d_{\text{synth}}), k_2) \quad (4)$$

The two-pass mechanism proves particularly valuable for multi-hop reasoning queries, where initial retrieval captures primary evidence and re-retrieval identifies the connective tissue linking disparate facts. Empirical evaluation (Section 6.3) demonstrates that this architecture yields substantial improvements on benchmarks requiring multi-step reasoning, such as HotpotQA and 2WikiMultiHop.

### 3.4 Context Filtering and Selection

Raw retrieval results typically contain redundancy and marginally relevant passages that, if passed directly to the generation model, would dilute the signal and potentially mislead response generation. HAH-RAG implements a principled filtering mechanism that balances two competing objectives, maximizing relevance to the query while ensuring diversity within the selected context set.

The intuition underlying this approach is straightforward. Relevance alone would favor passages that closely match the query, but multiple highly-similar passages provide diminishing returns, the

second document about a topic adds less information than the first. Diversity alone would favor coverage, but might include tangentially related content that confuses rather than informs. The optimal context set balances these considerations, providing comprehensive coverage of query-relevant information without redundancy.

We formalize this through a scoring function that combines relevance and diversity terms:

$$\text{Score}(c_i) = \alpha \cdot \text{Relevance}(c_i, q) + (1 - \alpha) \cdot \text{Diversity}(c_i, C) \quad (5)$$

where  $\alpha = 0.7$  by default, reflecting a preference for relevance while still rewarding diversity. The relevance component measures embedding similarity between the context passage and the query:

$$\text{Relevance}(c_i, q) = \frac{\mathbf{e}(c_i) \cdot \mathbf{e}(q)}{\|\mathbf{e}(c_i)\| \cdot \|\mathbf{e}(q)\|} \quad (6)$$

where  $\mathbf{e}(\cdot)$  denotes the embedding function. The diversity component penalizes passages that are too similar to others in the candidate set:

$$\text{Diversity}(c_i, C) = 1 - \frac{1}{|C| - 1} \sum_{j \neq i} \text{sim}(\mathbf{e}(c_i), \mathbf{e}(c_j)) \quad (7)$$

This formulation connects to the well-established Maximal Marginal Relevance (MMR) criterion [Carbonell and Goldstein \(1998\)](#), which has proven effective across information retrieval applications. The key insight is that optimal context selection is not simply a matter of ranking by relevance, but requires explicit consideration of information coverage and redundancy avoidance.

### 3.5 Notation Summary

Before proceeding to the detailed technical exposition, we establish notation that will be used throughout the remainder of this paper. Let  $D = \{d_1, d_2, \dots, d_N\}$  denote the document corpus containing  $N$  documents. Each document is segmented into chunks, yielding a chunk collection  $C = \{c_1, c_2, \dots, c_M\}$  where typically  $M \gg N$ . The set of retrievers operating in parallel is denoted  $R = \{r_1, r_2\}$ , where  $r_1$  represents the BM25 sparse retriever and  $r_2$  the dense vector retriever.

For reasoning classification, we denote the set of reasoning types as  $T = \{t_F, t_A, t_C, t_{CA}, t_H\}$ , corresponding to factual, analytical, comparative, causal, and hypothetical reasoning respectively. The reasoning classifier  $\phi : Q \rightarrow T \times [0, 1]$  maps queries to reasoning types with associated confidence scores. The embedding function  $\mathbf{e} : \Sigma^* \rightarrow \mathbb{R}^d$  maps text strings to  $d$ -dimensional vectors, where  $d = 768$  for the default embedding model.

Throughout, we use lowercase letters for scalars for instance,  $k$  for retrieval count, bold lowercase for vectors (e.g.,  $\mathbf{e}(q)$  for query embeddings), and uppercase letters for sets (e.g.,  $C$  for context set). Subscripts index into collections (for instance,  $c_i$  is the  $i$ -th context), while superscripts denote iteration or pass number (for instance,  $C^{(1)}$  for first-pass contexts).

## 4 Technical Architecture

This section details the core technical components of Template-based reasoning HAH-RAG with the Bayesian reasoning classifier that drives adaptive behavior, the hybrid retrieval system that combines sparse and dense methods, and the reranking and compression pipeline that refines retrieval results. Implementation details and hyperparameter specifications are provided in [Appendix C](#).

### 4.1 Bayesian Reasoning Detection

The reasoning detection subsystem must solve a classification problem, given a query  $q$ , determine which reasoning type  $t \in T$  best characterizes the cognitive demands of the question. This classification drives all downstream adaptations, retrieval parameters, fusion strategies, and generation prompts, making accuracy critical to system performance.

We formulate this as probabilistic inference using Bayes' theorem. Given a query  $q$ , we compute the posterior probability of each reasoning type:

$$P(t|q) = \frac{P(q|t) \cdot P(t)}{\sum_{t' \in T} P(q|t') \cdot P(t')} \quad (8)$$

The prior  $P(t)$  is initialized uniformly across reasoning types, reflecting absence of domain-specific knowledge about query distributions. In deployment scenarios where query statistics become available, these priors could be updated to reflect observed distributions.

The likelihood  $P(q|t)$  captures how well a query matches the expected characteristics of each reasoning type. Rather than model this likelihood directly, we approximate it through a weighted combination of three complementary evidence signals, pattern entropy, semantic coherence, and pattern coverage. The pattern entropy score measures the diversity of reasoning-specific linguistic markers found in the query, computed using Shannon entropy over matched patterns. Higher entropy indicates that multiple distinctive patterns are present, suggesting stronger evidence for the reasoning type. The semantic coherence score quantifies alignment between the query’s embedding and a canonical description of each reasoning type, capturing semantic similarity even when exact pattern matches are absent. The pattern coverage score computes the proportion of type-specific patterns that appear in the query, rewarding queries that exhibit multiple indicators of a reasoning type.

These three scores are combined through a weighted sum, with weights calibrated for each reasoning type to reflect the relative informativeness of different evidence sources. Factual reasoning, for instance, relies heavily on distinctive interrogative words (“what,” “who,” “when”), making pattern-based evidence highly informative. Hypothetical reasoning, in contrast, is characterized more by conditional constructions and speculative phrasing that may not match discrete patterns, leading to greater reliance on semantic coherence. The weighted evidence is further scaled by type-specific factors that encode prior knowledge about detection reliability. The complete specification of weights, thresholds, and pattern sets appears in Appendix C.1.

The Bayesian classifier achieves two important properties. First, it provides not just a classification but a confidence score, enabling downstream components to modulate their behavior based on classification certainty. When confidence is low, the system can adopt more conservative retrieval strategies that perform well across reasoning types. Second, the probabilistic formulation provides a principled framework for incorporating additional evidence sources or updating priors based on deployment experience.

## 4.2 Hybrid Retrieval with Adaptive Fusion

The retrieval component of HAH-RAG combines two complementary approaches, sparse lexical retrieval using BM25 and dense semantic retrieval using neural embeddings. This hybrid design reflects a fundamental insight from information retrieval research, lexical and semantic methods exhibit different strengths and failure modes, and their combination often outperforms either approach in isolation [Ma et al. \(2023\)](#); [Gao et al. \(2023b\)](#).

Sparse retrieval excels when queries contain distinctive terms that appear verbatim in relevant documents. A query for “CRISPR gene editing” benefits enormously from exact term matching, as the technical terminology provides strong relevance signals. Dense retrieval, in contrast, captures semantic relationships even when surface forms differ. A query about “methods to modify DNA” should retrieve documents about gene editing even without exact term overlap. By combining both approaches, HAH-RAG achieves robust retrieval across the spectrum of query types.

The sparse component implements BM25 [Robertson et al. \(2009\)](#), a probabilistic ranking function that scores documents based on term frequency, inverse document frequency, and document length normalization:

$$S_{\text{sparse}}(q, d) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1 \cdot (1 - b + b \cdot |d|/\text{avgdl})} \quad (9)$$

where  $f(q_i, d)$  is the frequency of term  $q_i$  in document  $d$ , and the IDF component weights terms by their discriminative power across the corpus. The hyperparameters  $k_1 = 1.5$  and  $b = 0.75$  control term frequency saturation and length normalization.

The dense component employs a bi-encoder architecture where queries and documents are independently encoded into vector representations using a transformer-based embedding model. Relevance is computed as cosine similarity between embeddings:

$$S_{\text{dense}}(q, d) = \frac{\mathbf{e}(q) \cdot \mathbf{e}(d)}{\|\mathbf{e}(q)\| \cdot \|\mathbf{e}(d)\|} \quad (10)$$

Document embeddings are precomputed during indexing and stored in a FAISS index [?](#) for efficient approximate nearest neighbor search.

The critical challenge in hybrid retrieval is score fusion, how should results from sparse and dense retrievers be combined into a unified ranking? This is non-trivial because the two scoring functions operate on different scales and have different statistical properties. HAH-RAG addresses this through Reciprocal Rank Fusion (RRF), which combines rankings rather than raw scores:

$$S_{\text{RRF}}(d) = \frac{1}{\beta + \text{rank}_{\text{sparse}}(d)} + \frac{1}{\beta + \text{rank}_{\text{dense}}(d)} \quad (11)$$

where  $\beta = 60$  controls the influence of rank positions. Documents appearing in both result lists receive contributions from both terms, while documents appearing in only one list are penalized for their absence in the other.

Beyond RRF, HAH-RAG supports adaptive fusion strategies that adjust retriever weights based on query characteristics. Queries containing proper nouns or technical terminology receive increased sparse weight, reflecting the importance of exact matching for such terms. Complex semantic queries receive increased dense weight to capture conceptual relationships. The system analyzes each query and selects an appropriate fusion strategy automatically. The complete specification of fusion methods and selection criteria appears in [Appendix C.2](#).

### 4.3 Cross-Encoder Reranking

Initial retrieval produces a candidate set that, while relevant, may not reflect optimal ordering for the specific query at hand. The bi-encoder architecture used in dense retrieval encodes queries and documents independently, preventing fine-grained interaction modeling between query terms and document content. To address this limitation, HAH-RAG employs a cross-encoder reranking stage that directly models query-document interactions.

Cross-encoders differ fundamentally from bi-encoders in their processing approach. Rather than encoding query and document separately, a cross-encoder processes the concatenated input through a single transformer, allowing attention mechanisms to capture nuanced relationships between query terms and document content. This architectural choice enables more accurate relevance assessment at the cost of increased computational expense, each query-document pair requires a forward pass through the model.

The reranking score for a query-document pair is computed as:

$$S_{\text{rerank}}(q, d) = \sigma(f([q; d])) \quad (12)$$

where  $f$  represents the cross-encoder transformer,  $[q; d]$  denotes the concatenated input with appropriate separator tokens, and  $\sigma$  is the sigmoid function normalizing outputs to the  $[0, 1]$  interval. We employ MS-MARCO-MiniLM as the default reranker, chosen for its balance between accuracy and inference speed.

Reranked results undergo threshold filtering to exclude passages with low relevance scores. Documents with reranking scores below the threshold (default 0.5) are removed from consideration, with a fallback to unfiltered results when filtering would produce an empty set. This filtering eliminates marginally relevant content that might otherwise dilute the context provided to the generation model. Implementation details including batch processing strategies and memory management appear in [Appendix C.3](#).

### 4.4 Contextual Compression

Even after reranking, the selected context set may contain redundant information or passages that exceed the language model’s context window. HAH-RAG addresses this through a contextual compression stage that produces compact, high-quality context sets while preserving essential information.

The compression pipeline chains three operations, initial retrieval returns an expanded candidate set (typically  $2k$  documents for a target of  $k$ ), reranking reorders these candidates by relevance, and compression selects the top fraction for final inclusion. Formally, given a query  $q$  and target context

count  $k$ :

$$\begin{aligned} C_{\text{candidates}} &= \text{Retrieve}(q, 2k) \\ C_{\text{reranked}} &= \text{Rerank}(q, C_{\text{candidates}}) \\ C_{\text{final}} &= \text{Top}(C_{\text{reranked}}, \lfloor |C_{\text{reranked}}| \cdot r_c \rfloor \end{aligned} \quad (13)$$

where  $r_c = 0.7$  is the compression ratio, retaining approximately 70% of reranked candidates.

The compression ratio represents a design trade-off between context comprehensiveness and noise reduction. Higher ratios preserve more potentially relevant content but risk including marginally useful passages. Lower ratios produce more focused context but may exclude useful information. Empirical evaluation (Section 6.3) suggests that 0.7 provides a robust default across diverse query types, though domain-specific applications might benefit from tuning.

#### 4.5 Asynchronous Processing Architecture

A critical design consideration for production RAG systems is latency. Users expect responsive interactions, yet the retrieval-rerank-generate pipeline involves multiple computationally expensive operations. HAH-RAG addresses this through an asynchronous processing architecture that parallelizes independent operations, substantially reducing end-to-end latency.

The key insight is that sparse and dense retrieval operations are independent, neither requires the output of the other, and both can execute concurrently. Similarly, the two-pass retrieval mechanism generates multiple search tasks that can execute in parallel. By exploiting this independence, HAH-RAG achieves latency approaching the maximum of individual operation times rather than their sum.

The sparse and dense retrievers execute concurrently, with results merged upon completion:

$$(D_{\text{sparse}}, D_{\text{dense}}) = \text{Parallel}(\text{BM25}(q, k), \text{Dense}(q, k)) \quad (14)$$

For the second retrieval pass, document chunks are submitted to parallel search operations, with results merged and deduplicated. This parallelization proves particularly valuable for complex queries requiring extensive context gathering, where sequential processing would impose unacceptable latency.

The asynchronous architecture extends beyond retrieval to encompass reranking and compression operations. When GPU resources are available, batch processing maximizes throughput; when resources are constrained, the system gracefully degrades to sequential processing with smaller batch sizes. This adaptive approach ensures robust performance across diverse deployment environments. Additional implementation details appear in Appendix C.4.

#### 4.6 Adaptive Generation Parameters

The generation phase presents a subtle optimization challenge, given retrieved context and a query, how should generation parameters be configured to produce optimal responses? HAH-RAG addresses this through adaptive parameter selection based on context characteristics.

A key consideration is token allocation. Language models have fixed context windows, and the available generation budget depends on how much of that window is consumed by the input (retrieved context plus query). Short contexts leave substantial room for extended generation; long contexts constrain the output length. HAH-RAG implements a context-ratio based allocation that dynamically adjusts the maximum generation length:

$$T_{\text{max}} = g\left(\frac{|C| + |q|}{L_{\text{model}}}\right) \quad (15)$$

where  $|C|$  is the context length in tokens,  $|q|$  is the query length,  $L_{\text{model}}$  is the model’s context window, and  $g$  is a piecewise function that allocates generation budget based on remaining capacity. The specific allocation schedule appears in Appendix C.5.

Beyond token allocation, HAH-RAG adapts frequency penalties based on context length. Longer contexts increase the risk of the model verbatim repeating retrieved content rather than synthesizing a coherent response. Higher frequency penalties for longer contexts discourage such repetition, encouraging the model to paraphrase and integrate information rather than copy it directly.

#### 4.7 Reasoning-Aware Prompt Templates

The final component of HAH-RAG’s adaptive architecture is reasoning-specific prompt engineering. Different reasoning types benefit from different generation guidance, factual queries call for precise, concise responses grounded in retrieved evidence; analytical queries benefit from structured decomposition and synthesis; causal queries require explicit reasoning about cause-effect relationships.

HAH-RAG implements this through a library of prompt templates, each designed for a specific reasoning type. The templates share a common structure, role specification, analysis guidance, context presentation, and query, but differ in the specific instructions provided to the model. Factual prompts emphasize precision and evidence grounding. Analytical prompts instruct the model to decompose problems and examine relationships. Comparative prompts guide structured evaluation of alternatives. Causal prompts direct attention to cause-effect chains. Hypothetical prompts encourage exploration of conditional scenarios.

The reasoning classifier (Section 4.1) determines which prompt template to instantiate for each query. When classification confidence is low, the system defaults to a general-purpose analytical prompt that performs acceptably across reasoning types. The complete prompt template specifications appear in Appendix C.6.

#### 4.8 Conversation Memory

While the preceding discussion has focused on single-turn interactions, practical applications often involve multi-turn conversations where context from previous exchanges informs subsequent queries. HAH-RAG maintains a sliding-window conversation buffer that preserves recent interaction history while bounding memory growth.

The conversation memory stores the most recent exchanges (user queries and assistant responses), providing this history as additional context for subsequent queries. This enables coherent multi-turn interactions where follow-up questions can reference previous discussion without explicit repetition. The buffer implements a fixed-size policy, when capacity is exceeded, oldest exchanges are discarded to make room for new ones.

Conversation history is formatted with reasoning-type-specific prefixes that cue the language model about the nature of prior context. A factual discussion receives a different framing than a causal analysis, helping the model maintain appropriate reasoning style across turns. This formatting integrates with the prompt template system, ensuring that conversation history is presented in a manner consistent with the current query’s reasoning requirements.

#### 4.9 Diversity-Aware Selection and Caching

Two additional mechanisms contribute to HAH-RAG’s effectiveness, diversity-aware document selection and result caching.

The diversity-aware selection mechanism addresses a common failure mode in retrieval systems, when multiple highly-similar documents are retrieved, they provide redundant information while consuming valuable context window space. HAH-RAG implements Maximal Marginal Relevance (MMR) Carbonell and Goldstein (1998), which balances relevance against redundancy by penalizing documents that are too similar to already-selected content. The MMR score for a candidate document considers both its relevance to the query and its dissimilarity from previously selected documents:

$$\text{MMR}(d) = \lambda \cdot \text{sim}(d, q) - (1 - \lambda) \cdot \max_{d' \in S} \text{sim}(d, d') \quad (16)$$

where  $S$  is the set of already-selected documents and  $\lambda$  controls the relevance-diversity trade-off.

Result caching addresses a different concern, computational efficiency for repeated or similar queries. In multi-turn conversations, follow-up queries often share substantial overlap with previous queries, and recomputing retrieval results from scratch is wasteful. HAH-RAG maintains an LRU (Least Recently Used) cache of retrieval results, keyed by query prefix and retrieval parameters. Cache hits bypass the expensive retrieval pipeline entirely, returning previously computed results. The cache maintains bounded size through eviction of least-recently-accessed entries. This strategy proves particularly effective for conversational workloads where query patterns exhibit temporal locality.

## 5 Evaluation Framework and Metrics

This section details the comprehensive evaluation framework employed to assess HAH-RAG’s performance, including generation quality metrics, retrieval effectiveness measures, and environmental impact quantification.

### 5.1 Generation Quality Metrics

For evaluating HAH-RAG’s performance across the selected datasets, we employ a comprehensive set of metrics that target different aspects of the proposed scheme against existing methods [Järvelin and Kekäläinen \(2002\)](#); [Barzilay and Lapata \(2008\)](#); [Maynez et al. \(2020\)](#); [Gehrmann et al. \(2022\)](#); [Wang et al. \(2018\)](#).

**NDCG Computation** The Normalized Discounted Cumulative Gain is computed as:

$$NDCG = \frac{DCG}{IDCG} \quad (17)$$

where:

$$DCG = \sum_{i=1}^k \frac{2^{rel_i} - 1}{\log_2(i + 1)} \quad (18)$$

and

$$IDCG = \sum_{i=1}^k \frac{2^{rel_i^*} - 1}{\log_2(i + 1)} \quad (19)$$

with  $rel_i$  representing relevance scores computed by cosine similarity between the retrieved and reference documents [Järvelin and Kekäläinen \(2002\)](#).

**Latency** System latency is measured as:

$$Latency = t_{end} - t_{start} + \sum_{i=1}^n CL_i \quad (20)$$

where  $CL_i$ , denoting cache lookup, represents the time spent checking the cache for existing results.

**Fluency** measures the grammatical correctness and natural flow of the generated text [Barzilay and Lapata \(2008\)](#). A perplexity-based fluency is adopted to compute the fluency, thus:

$$F = \exp \left( \frac{1}{N} \sum_{i=1}^N \log P(w_i | w_{i-n+1}, \dots, w_{i-1}) \right) \quad (21)$$

where  $P(w_i | context)$  is the probability of word  $w_i$  given its context, and  $N$  is the number of words.

**Coherence** evaluates the logical connection between sentences using cosine similarity of sentence embeddings, E [Barzilay and Lapata \(2008\)](#):

$$\text{Coherence, } c = \frac{1}{n-1} \sum_{i=1}^{n-1} \cos(E(s_i), E(s_{i+1})) \quad (22)$$

where  $E(s_i)$  represents the embedding of sentence  $i$ . The responses close to 1 are very coherent, and those close to 0 lack coherence.

**Relevance** measures the semantic similarity between the question and generated response [Maynez et al. \(2020\)](#):

$$\text{Relevance, } r = \cos(E(Q), E(A)) \quad (23)$$

where  $Q$  and  $A$  represent the question and answer respectively. Higher values indicate the correctness of the response with the question.

**Factuality** assesses the factual alignment between the generated text and source documents [Kryscinski et al. \(2020\)](#):

$$\text{Factuality}, f = \max_i \cos(E(\text{generated}), E(\text{source}_i)) \quad (24)$$

Higher factuality score is preferred with scores ranging between  $[0, 1]$ .

**Advanced HHEM** incorporates coherence and relevance:

$$\text{Adv HHEM} = \frac{\bar{s} \cdot f \cdot c \cdot r}{1 + \bar{s} \cdot f} \quad (25)$$

where  $c$  is coherence score and  $r$  is relevance score. It rewards outputs that are not only factual but also coherent and relevant. Hence, higher values are preferred.

**Correctness** combines semantic similarity with key information overlap [Wang et al. \(2018\)](#); [Ribeiro et al. \(2020\)](#):

$$\text{Correctness} = 0.5 \cdot ssc + 0.5 \cdot kis \quad (26)$$

where  $ssc$  is the semantic score and  $kis$  is the key info score and computed from the overlap of n-grams between generated and reference texts.

The metrics are scaled using a constant factor (scale metric) for standardization across different evaluation scenarios. Each metric produces scores in the range  $[0, 1]$ , where higher values generally indicate better performance, except for Advanced HHEM, where lower values suggest less hallucination.

## 5.2 Energy Consumption and Environmental Impact

The computation of energy consumption and environmental impact metrics is based on several key components [Henderson et al. \(2020\)](#); [Strubell et al. \(2019\)](#):

### 5.2.1 Power Consumption

**CPU Power** [Patterson et al. \(2022\)](#) is estimated based on CPU frequency and utilization:

$$P_{\text{CPU}} = P_{\text{base}} \cdot \frac{f_{\text{current}}}{f_{\text{max}}} \cdot \text{CPU}_{\text{util}} \quad (27)$$

where  $P_{\text{base}}$  is the base CPU power (typically 65W),  $f_{\text{current}}$  and  $f_{\text{max}}$  are current and maximum CPU frequencies, and  $\text{CPU}_{\text{util}}$  is the CPU utilization percentage.

**GPU Power** is computed either through direct measurement or estimation:

$$P_{\text{GPU}} = \begin{cases} P_{\text{measured}} & \text{if available} \\ 250W \cdot \text{GPU}_{\text{util}} & \text{otherwise} \end{cases} \quad (28)$$

where  $\text{GPU}_{\text{util}}$  is the GPU utilization percentage.

### 5.2.2 Energy Consumption

The total energy consumption is calculated by integrating power over time:

$$E_{\text{total}} = \sum_t \frac{(P_{\text{CPU}} + P_{\text{GPU}}) \cdot \Delta t}{1000 \cdot 3600} \text{ kWh} \quad (29)$$

where  $\Delta t$  is the sampling interval (typically 0.1 seconds).

### 5.2.3 CO<sub>2</sub> Emissions

CO<sub>2</sub> emission [Lacoste et al. \(2019\)](#) is computed using country-specific carbon intensity factors:

$$\text{CO}_2 = \frac{E_{\text{total}} \cdot CI_{\text{country}}}{1000} \text{ kg} \quad (30)$$

where  $CI_{\text{country}}$  is the carbon intensity factor (in g CO<sub>2</sub>/kWh) for the specific country. Since this experiment is conducted in France, we use a CI of 70 g CO<sub>2</sub> / kWh and a global average CI of 475 g CO<sub>2</sub>/kWh.

### 5.2.4 Resource Metrics

The average resource utilization metrics are computed during the monitoring period.

$$\text{CPU}_{\text{avg}} = \frac{1}{N} \sum_{t=1}^N \text{CPU}_{\text{util}}(t) \quad (31)$$

$$\text{GPU}_{\text{avg}} = \frac{1}{N} \sum_{t=1}^N \text{GPU}_{\text{util}}(t) \quad (32)$$

$$\text{Memory}_{\text{avg}} = \frac{1}{N} \sum_{t=1}^N \text{Memory}_{\text{util}}(t) \quad (33)$$

where  $N$  is the total number of observations collected during the monitoring period.

These metrics are continuously monitored through a dedicated thread with a sampling interval of 0.1 seconds (sleep time), providing detailed resource usage profiles throughout the RAG execution time. The measurements account for both computational resources (CPU, GPU) and their environmental impact through energy consumption and CO<sub>2</sub> emissions calculations.

## 6 Experimentation

### 6.1 Experimental Setup

All experiments were conducted on a single NVIDIA GeForce RTX 5090 with 32GB VRAM, paired with a 60-core CPU and CUDA 12.4. Table 2 summarizes the language models evaluated, and Table 4 describes the benchmark datasets.

Table 2: Language models evaluated in our experiments. All models use W4A16 quantization (4-bit weights, 16-bit activations) unless otherwise noted. Parameter counts refer to the base (unquantized) model, all served using vLLM. The context length column reports the experimental setting, not the original model capacity.

| Model                                    | Base Architecture         | Params | Quantization | Exp. Context | Original Context |
|------------------------------------------|---------------------------|--------|--------------|--------------|------------------|
| Llama-3.2-3B (Grattafiori et al., 2024b) | Meta Llama 3.2            | 3B     | W4A16        | 8K           | 128K             |
| DeepSeek-R1-3B (Guo et al., 2025)        | DeepSeek-R1-Distill-Llama | 3B     | None         | 8K           | 64K              |
| Gemma-3-4B (Team et al., 2025)           | Google Gemma 3            | 4B     | W4A16        | 8K           | 128K             |
| Phi-3-mini-4K (Abdin et al., 2024)       | Microsoft Phi-3           | 3.8B   | None         | 4K           | 4K               |
| Llama-3-8B (Grattafiori et al., 2024a)   | Meta Llama 3              | 8B     | W4A16        | 8K           | 8K               |
| DeepSeek-R1-8B (Guo et al., 2025)        | DeepSeek-R1-Distill-Llama | 8B     | W4A16        | 8K           | 64K              |
| Gemma-2-9B (Team et al., 2024)           | Google Gemma 2            | 9B     | W4A16        | 8K           | 8K               |
| Gemma-3-12B (Team et al., 2025)          | Google Gemma 3            | 12B    | W4A16        | 8K           | 128K             |

The model selection in Table 2 reflects a deliberate strategy to evaluate HAH-RAG across diverse architectural families and parameter scales. We include models from four major vendors, Meta (Llama), Google (Gemma), Microsoft (Phi), and DeepSeek, to assess generalization across different pretraining objectives and instruction-tuning approaches. The parameter range spans 3B to 12B, enabling analysis of how retrieval architecture benefits vary with model capacity. Notably, the DeepSeek-R1 models are distilled from reasoning-specialized teachers, providing insight into whether HAH-RAG’s reasoning detection complements or conflicts with inherent reasoning capabilities. W4A16 quantization (4-bit weights, 16-bit activations) is employed for most models to enable practical deployment on consumer-grade hardware while maintaining competitive quality. All models are served through vLLM (Kwon et al., 2023) for efficient batched inference.

It is important to distinguish between the experimental context length and the original model capacity. The **Exp. Context** column indicates the maximum context window configured during our experiments (uniformly 8K tokens, except for Phi-3-mini-4K at 4K), while the **Original Context** column reports the native context length supported by each model architecture. Several models, including Llama-3.2, Gemma-3, and DeepSeek-R1 variants, support substantially longer contexts

(64K–128K tokens) in their original configurations. We constrain the experimental context to 8K for two reasons: first, to ensure fair comparison across models with heterogeneous native capacities; second, because RAG systems typically operate with retrieved contexts well below 8K tokens, making larger windows unnecessary for our evaluation scenarios. This standardization isolates the effect of retrieval architecture from confounding differences in context utilization capabilities.

Table 3: Key hyperparameters and system configurations used across all experiments.

| Component  | Parameter                     | Value                       |
|------------|-------------------------------|-----------------------------|
| Retrieval  | Embedding Model               | all-mpnet-base-v2 (768-dim) |
|            | Vector Store                  | FAISS (inner product)       |
|            | Default $k$ (retrieval count) | 5 (scaled for large models) |
| Reranking  | Cross-Encoder                 | ms-marco-MiniLM-L-12-v2     |
|            | Relevance Threshold           | 0.5                         |
|            | Compression Ratio             | 0.7                         |
| Fusion     | Default Method                | Reciprocal Rank Fusion      |
|            | RRF Constant ( $\beta$ )      | 60                          |
| Generation | Temperature                   | 0.7                         |
|            | Max New Tokens                | Adaptive (2048–8192)        |
|            | Frequency Penalty             | Adaptive (0.01–0.10)        |
| Hardware   | GPU                           | NVIDIA RTX 5090 (32GB VRAM) |
|            | CPU                           | 60 cores                    |
|            | CUDA Version                  | 12.4                        |

The Table 3 presents the hyperparameter configuration held constant across all experiments. The retrieval component uses sentence-transformers’ **all-mpnet-base-v2** embeddings, chosen for their strong performance on semantic similarity benchmarks. FAISS serves as the vector store backend, configured for inner product search after L2 normalization. The default retrieval count  $k = 5$  balances context comprehensiveness against noise introduction, with automatic scaling for models with extended context windows. The reranking stage employs **ms-marco-MiniLM-L-12-v2**, a cross-encoder trained on the MS-MARCO passage ranking dataset, with a relevance threshold of 0.5 filtering low-confidence retrievals. The compression ratio of 0.7 retains the top 70% of reranked passages, empirically determined to preserve essential information while reducing redundancy. Generation parameters follow standard practices: temperature 0.7 encourages diverse but coherent outputs, while adaptive token limits and frequency penalties respond to input context length as described in Section 4.

## 6.2 Datasets

The evaluation of HAH-RAG follows a comprehensive scheme across multiple dimensions of performance using established benchmarks. To assess basic retrieval performance and knowledge integration, we employ three primary datasets: Natural Questions (NQ) (Kwiatkowski et al., 2019), which provides real user queries from Google Search containing 307K training examples and 8K development examples; and TriviaQA (Joshi et al., 2017), which offers a large-scale question answering dataset with evidence documents comprising 95K questions with answers.

Table 4: Benchmark datasets used for evaluation, spanning factual QA, commonsense reasoning, multi-hop reasoning, and ambiguous queries.

| Dataset       | Task Type             | Size  | Reasoning Required |
|---------------|-----------------------|-------|--------------------|
| SQuAD         | Extractive QA         | 100K+ | Single-hop         |
| CommonsenseQA | Commonsense Reasoning | 12K   | Single-hop         |
| HotpotQA      | Multi-hop QA          | 113K  | Multi-hop          |
| 2WikiMultiHop | Multi-hop QA          | 193K  | Multi-hop          |
| AmbigQA       | Ambiguous QA          | 14K   | Disambiguation     |

The dataset selection in Table 4 is designed to stress-test different aspects of HAH-RAG’s architecture. SQuAD and CommonsenseQA represent single-hop reasoning tasks where the answer can typically be found in a single retrieved passage, these serve as baselines where simpler retrieval should suffice. HotpotQA and 2WikiMultiHop require multi-hop reasoning, where evidence must be gathered from multiple documents and synthesized, precisely the scenario for which HAH-RAG’s two-pass retrieval and context filtering mechanisms are designed. AmbigQA introduces ambiguity, where queries admit multiple valid interpretations requiring the system to either disambiguate or provide comprehensive coverage. This diversity enables fine-grained analysis of when and why HAH-RAG’s additional complexity yields benefits.

To evaluate the capability of HAH-RAG in handling complex reasoning and multistep retrieval, we use HotpotQA (Yang et al., 2018). HotpotQA is specifically selected to assess the effectiveness of our hyper-layered asynchronous scheme in managing multi-hop questions and comprises 113K Wikipedia-based question-and-answer pairs. For comprehensive response generation and context integration, we employ NarrativeQA (Kočíský et al., 2018), which is particularly relevant to evaluate how HAH-RAG handles chunk processing and maintains context coherence in longer responses.

### 6.3 Results

We organise our empirical investigation around a central question, does the additional architectural complexity of HAH-RAG translate into meaningful improvements in generation quality, and if so, under what conditions? To address this, we conduct experiments across five established question-answering benchmarks, namely SQuAD, CommonsenseQA, HotpotQA, 2WikiMultiHop, and AmbigQA. These benchmarks were selected to span a range of cognitive demands, from extractive retrieval to multi-hop reasoning to disambiguation under ambiguity. We compare five pipeline configurations, Naive (relying solely on dense vector retrieval), Hybrid (combining sparse BM25 with dense retrieval through reciprocal rank fusion), HAH (our full hyper-layered asynchronous hybrid architecture), Reasoning (HAH augmented with Bayesian reasoning detection), and Mini-Reasoning (a lightweight variant thereof). Eight language models, spanning three billion to twelve billion parameters, serve as the generative backbone for each configuration, allowing us to examine how retrieval architecture interacts with model capacity.

### 6.4 Quantitative Analysis

#### 6.4.1 Generation Quality Metrics

We begin by examining the core question of whether retrieval architecture affects the quality of generated responses. The Table 5 presents our comprehensive findings across all model and pipeline combinations. The patterns that emerge from this analysis prove both consistent and instructive.

**Factuality and Relevance.** Perhaps the most striking finding concerns factuality. We observe that HAH consistently outperforms baseline configurations across the model spectrum, with improvements that, while varying in magnitude, prove remarkably robust. Consider DeepSeek-R1-3B as an illustrative case, under HAH, this model achieves a factuality score of 0.775 compared to 0.728 for Naive and 0.730 for Hybrid, representing an improvement of approximately 6% (see Table 5). This improvement, though modest in absolute terms, represents a meaningful reduction in hallucinated or unsupported content.

The pattern extends beyond individual models. Phi-3-mini-4K, when paired with HAH, attains the highest factuality score observed in our experiments at 0.817, substantially exceeding both Naive (0.767) and Hybrid (0.731) configurations (see Figure 4 for a visual comparison across all models). We interpret this finding as evidence that the multi-layer retrieval strategy, combined with contextual compression, succeeds in its intended purpose, filtering noise from retrieved context whilst preserving semantically relevant material. The compression mechanism appears to act as a form of attention guidance, directing the generative model toward information most pertinent to the query at hand.

Relevance scores tell a complementary story. Here, we find not merely incremental improvements but, in certain cases, dramatic recovery from baseline failures. Llama-3-8B provides a particularly instructive example. Under the Hybrid configuration, this model exhibits a relevance score of only 0.314; under HAH, performance recovers to 0.764, more than doubling the Hybrid result. This observation merits careful consideration. Simple fusion strategies, whilst theoretically sound, may in-

introduce instabilities that manifest as inconsistent retrieval quality. The reranking mechanism central to HAH appears to mitigate these instabilities, providing a more reliable foundation for downstream generation.

**Fluency and Coherence Trade-offs.** Our analysis reveals an intriguing tension between fluency and coherence that warrants discussion. Several models achieve near-ceiling fluency scores under HAH; DeepSeek-R1-Distill-Llama-8B, for instance, reaches 0.981 fluency, suggesting that the compressed context does not impede the mechanical quality of text generation. However, we note that coherence, as measured by our evaluation framework, sometimes decreases relative to Naive baselines.

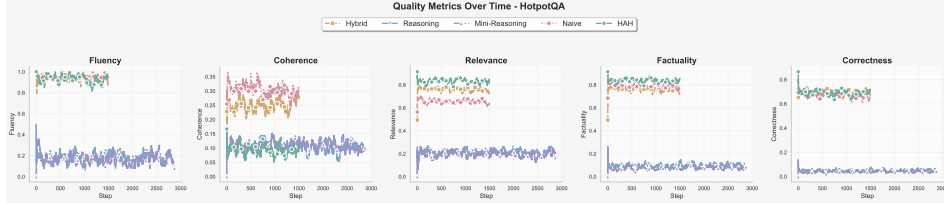


Figure 3: Evolution of quality metrics over evaluation instances on HotpotQA. The smoothed trajectories (rolling window) reveal that HAH maintains stable, high-quality outputs throughout evaluation except for coherence; whereas Reasoning variants exhibit greater variance, consistent with their bypass mechanism triggering inconsistently on benchmark queries.

This trade-off appears model-dependent rather than architectural, as illustrated in Figure 6 which tracks quality metrics over evaluation instances. Certain model families, particularly those with strong instruction-following capabilities, maintain robust coherence alongside high fluency. Others exhibit what might be characterised as a “compression sensitivity,” wherein the distillation of context disrupts inter-sentence flow patterns upon which their generation relies. We suspect that future work on adaptive compression, tuned to model-specific requirements, could ameliorate this limitation.

**Correctness and Overall Performance.** The correctness metric provides perhaps the most holistic view of answer quality, combining semantic similarity with key information overlap. Examining this measure, we find that HAH ranks first or second across the majority of evaluated models (see Figure 5). DeepSeek-R1-Distill-Llama-8B achieves the highest correctness observed at 0.758. The architecture’s advantage proves most pronounced where it is most needed, for models that struggle under simpler retrieval paradigms. Llama-3-8B, to continue our earlier example, improves from 0.383 under Naive to 0.664 under HAH, representing a relative gain of approximately 73%. Such gains suggest that sophisticated retrieval architecture can compensate, at least partially, for limitations in model capacity or training.

#### 6.4.2 Resource Efficiency Analysis

A natural concern with any multi-layer architecture is computational overhead. Does HAH’s quality advantage come at prohibitive cost? We address this question through detailed resource profiling, presented in Table 6.

**Latency Characteristics.** We find that HAH achieves competitive, and often superior, latency relative to baseline configurations. This finding may seem counterintuitive given the additional retrieval passes involved, but the asynchronous architecture proves its worth here. For Gemma-3-4B, HAH achieves the lowest latency at 0.335s, outperforming both Naive (0.458s) and Hybrid (1.025s). By parallelising operations that would otherwise execute sequentially, HAH amortises the cost of its complexity.

The Hybrid pipeline, by contrast, exhibits notably higher latency for certain model configurations. This reflects the sequential nature of traditional sparse-dense fusion, where results from one retrieval system must be obtained before fusion can occur. HAH’s design circumvents this bottleneck through concurrent execution.

We note that the Reasoning and Mini-Reasoning variants present a distinct profile. These configurations sometimes achieve lower latency than HAH, but examination of quality metrics reveals a troubling pattern, the latency gains appear to stem from the trivial question bypass mechanism triggering inappropriately, leading to responses generated without proper contextual grounding. This observation has implications for deployment that we discuss further in our limitations section.

**GPU Utilisation and Power Consumption.** The asynchronous architecture translates to efficient hardware utilisation. Across multiple model configurations, HAH achieves the lowest GPU utilisation among quality-preserving pipelines. For DeepSeek-R1-3B, HAH records GPU utilisation of 1.70% and power consumption of approximately 4.2W; similar patterns hold for Gemma-3-4B (1.02% GPU) and Gemma-3-12B (consult Table 6 for complete values). These results validate our design hypothesis, parallelised retrieval with intelligent caching reduces redundant computation, leading to lower power draw and, consequently, reduced environmental impact.

**Economic Cost.** For deployment at scale, economic considerations inevitably influence architectural decisions. Our cost analysis reveals that HAH achieves cost parity or improvement relative to Naive retrieval for six of eight evaluated models, whilst delivering superior generation quality. The savings, whilst small on a per-query basis, compound significantly at scale. Processing one million queries with Gemma-2-9B costs approximately €162 under HAH versus €176 under Naive, representing an 8% reduction without sacrificing, and indeed whilst improving, response quality. Table 7 provides a detailed breakdown of latency, GPU utilisation, and cost across all dataset-model-pipeline combinations, enabling fine-grained analysis of where efficiency gains concentrate.

### 6.4.3 Dataset-Specific Performance

Having established aggregate patterns, we turn to the question of whether HAH’s advantages generalise across task types or concentrate in specific domains. The detailed per-dataset analysis (Table 8 in the Appendix) illuminates important interactions between dataset characteristics and pipeline effectiveness.

**Multi-hop Reasoning Tasks.** We find that HAH exhibits its strongest relative performance on benchmarks requiring multi-step reasoning. HotpotQA and 2WikiMultiHop, both designed to assess an agent’s capacity to synthesise evidence across multiple documents, reveal marked improvements under HAH. Phi-3-mini-4K on HotpotQA achieves factuality of 0.837 under HAH versus 0.761 under Hybrid, an improvement of approximately 10%.

We attribute this pattern to HAH’s two-pass retrieval mechanism. Questions requiring multi-hop reasoning often contain implicit sub-queries that a single retrieval pass may fail to fully address. By using initially retrieved context to inform subsequent retrieval, HAH effectively expands its semantic coverage, capturing bridging documents that connect disparate pieces of evidence. The Hybrid pipeline, lacking this re-querying capability, more frequently returns incomplete evidence chains.

**Commonsense and Ambiguous Queries.** CommonsenseQA and AmbigQA present qualitatively different challenges, requiring world knowledge integration and disambiguation respectively. We observe that HAH maintains robust performance on these benchmarks, though the margins of improvement narrow relative to multi-hop tasks. DeepSeek-R1-3B achieves relevance of 0.743 on CommonsenseQA under HAH compared to 0.676 for Naive, representing approximately 10% improvement. For AmbigQA, the contextual compression mechanism appears particularly beneficial; Phi-3-mini-4K reaches factuality of 0.811 under HAH, perhaps because compression forces the selection of maximally informative context, thereby reducing ambiguity in the material presented to the generative model.

**Extractive QA.** On SQuAD, representing traditional extractive question answering, we observe a narrowing of performance gaps between pipelines. This finding aligns with theoretical expectations, when answers are explicitly present in retrieved passages, sophisticated reranking provides diminishing returns. The answer span can be identified through surface matching as readily as through semantic analysis. Nevertheless, HAH maintains competitive performance; DeepSeek-R1-Distill-Llama-8B achieves correctness of 0.754 under HAH, the highest observed for this benchmark (see

Table 8 for model-specific results). We interpret this as evidence that HAH’s overhead, whilst not always necessary, is not prohibitive even for simpler tasks.

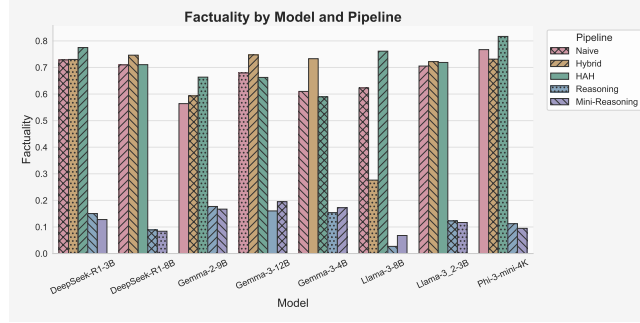


Figure 4: Factuality scores across models and pipelines. HAH achieves the highest or near-highest factuality for the majority of evaluated models, illustrating the effectiveness of multi-layer retrieval with contextual compression. Error bars, where present, indicate variance across dataset subsets.

Table 5: Comprehensive performance across datasets, averaged per model–pipeline pair. Up arrows indicate higher values are preferred; best results per model are shown in **bold**. Notably, the HAH pipeline consistently yields the strongest relevance and factuality scores for most models, indicating a more faithful generation profile.

| Model                        | Pipeline       | NDCG ↑       | R-1 ↑        | R-2 ↑        | R-L ↑        | Fluency ↑    | Coherence ↑  | Relevance ↑  | Factuality ↑ | Correctness ↑ |
|------------------------------|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|
| DeepSeek-R1-3B               | Naive          | <b>0.923</b> | <b>0.052</b> | <b>0.016</b> | <b>0.052</b> | 0.896        | 0.264        | 0.730        | 0.728        | 0.525         |
|                              | Hybrid         | <b>0.923</b> | 0.006        | 0.000        | 0.006        | 0.891        | 0.310        | 0.731        | 0.730        | 0.670         |
|                              | HAH            | <b>0.923</b> | 0.000        | 0.000        | 0.000        | <b>0.904</b> | <b>0.332</b> | <b>0.773</b> | <b>0.775</b> | <b>0.694</b>  |
|                              | Reasoning      | 0.424        | 0.012        | 0.005        | 0.011        | 0.389        | 0.135        | 0.318        | 0.150        | 0.105         |
|                              | Mini-Reasoning | 0.437        | 0.019        | 0.008        | 0.018        | 0.390        | 0.149        | 0.326        | 0.128        | 0.089         |
| DeepSeek-R1-Distill-Llama-8B | Naive          | <b>0.923</b> | <b>0.025</b> | <b>0.002</b> | <b>0.025</b> | 0.855        | 0.359        | 0.736        | 0.710        | 0.505         |
|                              | Hybrid         | <b>0.923</b> | 0.005        | 0.000        | 0.005        | 0.941        | <b>0.363</b> | <b>0.747</b> | <b>0.746</b> | 0.736         |
|                              | HAH            | <b>0.923</b> | 0.000        | 0.000        | 0.000        | <b>0.981</b> | 0.340        | 0.708        | 0.711        | <b>0.758</b>  |
|                              | Reasoning      | 0.424        | 0.002        | 0.001        | 0.002        | 0.153        | 0.056        | 0.140        | 0.089        | 0.061         |
|                              | Mini-Reasoning | 0.437        | 0.004        | 0.002        | 0.004        | 0.195        | 0.074        | 0.162        | 0.084        | 0.058         |
| Gemma-2-9B                   | Naive          | <b>0.923</b> | 0.038        | 0.000        | 0.038        | 0.622        | 0.103        | 0.618        | 0.564        | 0.326         |
|                              | Hybrid         | <b>0.923</b> | 0.012        | 0.000        | 0.012        | 0.609        | <b>0.141</b> | 0.603        | 0.593        | 0.489         |
|                              | HAH            | <b>0.923</b> | 0.000        | 0.000        | 0.000        | <b>0.686</b> | 0.046        | <b>0.664</b> | <b>0.664</b> | <b>0.561</b>  |
|                              | Reasoning      | 0.425        | <b>0.052</b> | <b>0.022</b> | <b>0.050</b> | 0.353        | 0.043        | 0.319        | 0.177        | 0.099         |
|                              | Mini-Reasoning | 0.437        | 0.030        | 0.013        | 0.028        | 0.411        | 0.076        | 0.339        | 0.167        | 0.106         |
| Gemma-3-12B                  | Naive          | <b>0.923</b> | <b>0.232</b> | <b>0.187</b> | <b>0.232</b> | 0.810        | 0.153        | 0.734        | 0.679        | 0.410         |
|                              | Hybrid         | <b>0.923</b> | 0.056        | 0.000        | 0.056        | <b>0.853</b> | <b>0.285</b> | <b>0.747</b> | <b>0.747</b> | <b>0.650</b>  |
|                              | HAH            | <b>0.923</b> | 0.000        | 0.000        | 0.000        | 0.764        | 0.055        | 0.659        | 0.663        | 0.589         |
|                              | Reasoning      | 0.425        | 0.076        | 0.042        | 0.075        | 0.345        | 0.038        | 0.269        | 0.160        | 0.090         |
|                              | Mini-Reasoning | 0.437        | 0.031        | 0.016        | 0.031        | 0.402        | 0.070        | 0.303        | 0.195        | 0.125         |
| Gemma-3-4B                   | Naive          | <b>0.923</b> | <b>0.073</b> | <b>0.047</b> | <b>0.073</b> | 0.683        | 0.146        | 0.650        | 0.610        | 0.361         |
|                              | Hybrid         | <b>0.923</b> | 0.000        | 0.000        | 0.000        | <b>0.803</b> | <b>0.283</b> | <b>0.733</b> | <b>0.733</b> | <b>0.615</b>  |
|                              | HAH            | <b>0.923</b> | 0.033        | 0.000        | 0.033        | 0.621        | 0.058        | 0.583        | 0.590        | 0.485         |
|                              | Reasoning      | 0.426        | 0.060        | 0.032        | 0.060        | 0.284        | 0.035        | 0.296        | 0.154        | 0.083         |
|                              | Mini-Reasoning | 0.437        | 0.021        | 0.011        | 0.021        | 0.417        | 0.084        | 0.327        | 0.172        | 0.113         |
| Llama-3-8B                   | Naive          | <b>0.923</b> | <b>0.071</b> | <b>0.047</b> | <b>0.071</b> | 0.746        | <b>0.193</b> | 0.674        | 0.623        | 0.383         |
|                              | Hybrid         | <b>0.923</b> | 0.000        | 0.000        | 0.000        | 0.334        | 0.104        | 0.314        | 0.276        | 0.269         |
|                              | HAH            | <b>0.923</b> | 0.024        | 0.000        | 0.024        | <b>0.848</b> | 0.157        | <b>0.764</b> | <b>0.761</b> | <b>0.664</b>  |
|                              | Reasoning      | 0.426        | 0.004        | 0.002        | 0.004        | 0.061        | 0.015        | 0.083        | 0.027        | 0.021         |
|                              | Mini-Reasoning | 0.437        | 0.008        | 0.004        | 0.008        | 0.119        | 0.024        | 0.127        | 0.068        | 0.045         |
| Phi-3-mini-4K                | Naive          | <b>0.923</b> | 0.002        | 0.000        | 0.002        | <b>0.901</b> | <b>0.337</b> | 0.662        | 0.767        | 0.653         |
|                              | Hybrid         | <b>0.923</b> | <b>0.031</b> | 0.000        | <b>0.031</b> | 0.868        | 0.293        | 0.732        | 0.731        | 0.656         |
|                              | HAH            | <b>0.923</b> | 0.000        | 0.000        | 0.000        | 0.891        | 0.088        | <b>0.816</b> | <b>0.817</b> | <b>0.696</b>  |
|                              | Reasoning      | 0.424        | 0.008        | <b>0.003</b> | 0.008        | 0.175        | 0.108        | 0.229        | 0.112        | 0.059         |
|                              | Mini-Reasoning | 0.437        | 0.002        | 0.001        | 0.002        | 0.174        | 0.108        | 0.217        | 0.095        | 0.050         |
| Llama-3-2-3B                 | Naive          | <b>0.923</b> | <b>0.049</b> | <b>0.020</b> | <b>0.049</b> | <b>0.882</b> | <b>0.404</b> | 0.712        | 0.705        | 0.498         |
|                              | Hybrid         | <b>0.923</b> | 0.000        | 0.000        | 0.000        | 0.838        | 0.351        | <b>0.722</b> | <b>0.722</b> | <b>0.639</b>  |
|                              | HAH            | <b>0.923</b> | 0.000        | 0.000        | 0.000        | 0.810        | 0.375        | 0.717        | 0.719        | 0.614         |
|                              | Reasoning      | 0.424        | 0.012        | 0.006        | 0.012        | 0.306        | 0.105        | 0.253        | 0.123        | 0.078         |
|                              | Mini-Reasoning | 0.437        | 0.010        | 0.004        | 0.010        | 0.265        | 0.106        | 0.205        | 0.117        | 0.082         |

## 6.5 Qualitative Analysis

The quantitative patterns documented above invite deeper examination. What architectural mechanisms account for the observed improvements, and why do certain configurations succeed where others fail? In this section, we move beyond aggregate statistics to develop an interpretive account of our findings.

Table 6: Resource Usage and Cost Metrics: values are averaged across datasets for each model-pipeline pair. Best (lowest) results per model are shown in **bold**.

| Model                        | Pipeline       | Latency (s) ↓ | CPU (%) ↓    | Memory (%) ↓ | GPU Util. (%) ↓ | GPU Power (W) ↓ | Total Cost (EUR) ↓ |
|------------------------------|----------------|---------------|--------------|--------------|-----------------|-----------------|--------------------|
| DeepSeek-R1-3B               | Naive          | 1.636         | 22.45        | <b>6.80</b>  | 1.79            | 4.5             | 0.001632           |
|                              | Hybrid         | 1.735         | 24.54        | 6.86         | 2.00            | 4.9             | 0.001787           |
|                              | HAH            | <b>1.509</b>  | 21.30        | 6.83         | <b>1.70</b>     | <b>4.2</b>      | <b>0.001354</b>    |
|                              | Reasoning      | 2.334         | 12.74        | 7.92         | 78.58           | 196.8           | 0.047131           |
|                              | Mini-Reasoning | 5.157         | <b>10.22</b> | 7.25         | 37.40           | 93.6            | 0.052820           |
| DeepSeek-R1-Distill-Llama-8B | Naive          | 2.072         | <b>9.28</b>  | 8.66         | <b>3.42</b>     | <b>8.6</b>      | 0.002602           |
|                              | Hybrid         | 3.215         | 11.14        | 8.82         | 4.03            | 10.1            | 0.003347           |
|                              | HAH            | 2.185         | 9.35         | 8.73         | 6.13            | 15.3            | <b>0.002582</b>    |
|                              | Reasoning      | <b>0.681</b>  | 20.17        | 8.67         | 54.43           | 136.7           | 0.010750           |
|                              | Mini-Reasoning | 1.973         | 16.50        | <b>8.02</b>  | 71.07           | 178.5           | 0.028561           |
| Gemma-2-9B                   | Naive          | 0.454         | 7.37         | 8.59         | 1.36            | 3.3             | 0.000176           |
|                              | Hybrid         | 0.584         | <b>7.28</b>  | <b>8.23</b>  | 1.26            | <b>3.1</b>      | 0.000220           |
|                              | HAH            | <b>0.406</b>  | 7.55         | 8.65         | <b>1.24</b>     | 3.2             | <b>0.000162</b>    |
|                              | Reasoning      | 0.427         | 34.70        | 9.15         | 31.96           | 79.9            | 0.004724           |
|                              | Mini-Reasoning | 0.855         | 32.48        | 8.67         | 19.87           | 50.1            | 0.006943           |
| Gemma-3-12B                  | Naive          | 0.599         | <b>7.81</b>  | 9.35         | <b>0.00</b>     | <b>0.0</b>      | 0.000115           |
|                              | Hybrid         | 1.171         | 12.44        | <b>5.66</b>  | 72.05           | 180.1           | 0.011536           |
|                              | HAH            | <b>0.409</b>  | 8.64         | 9.83         | <b>0.00</b>     | <b>0.0</b>      | <b>0.000098</b>    |
|                              | Reasoning      | 0.419         | 34.36        | 9.96         | 30.76           | 76.9            | 0.004528           |
|                              | Mini-Reasoning | 1.562         | 22.84        | 8.97         | 51.27           | 129.0           | 0.024101           |
| Gemma-3-4B                   | Naive          | 0.458         | 9.23         | 10.13        | 1.13            | 2.8             | 0.000200           |
|                              | Hybrid         | 1.025         | 21.76        | 9.73         | 1.36            | 3.4             | 0.000890           |
|                              | HAH            | <b>0.335</b>  | <b>7.53</b>  | 10.26        | <b>1.02</b>     | <b>2.6</b>      | <b>0.000132</b>    |
|                              | Reasoning      | 0.371         | 36.83        | 10.25        | 31.07           | 77.8            | 0.004049           |
|                              | Mini-Reasoning | 0.801         | 30.29        | <b>9.57</b>  | 38.31           | 95.2            | 0.010708           |
| Llama-3-8B                   | Naive          | 0.589         | 9.87         | 23.56        | 55.97           | 141.3           | 0.004412           |
|                              | Hybrid         | 0.459         | <b>7.50</b>  | 23.62        | 53.55           | 134.0           | 0.003249           |
|                              | HAH            | 0.479         | 8.08         | 23.54        | 55.59           | 139.1           | 0.003747           |
|                              | Reasoning      | 0.305         | 33.75        | <b>8.49</b>  | 16.30           | 40.6            | 0.002017           |
|                              | Mini-Reasoning | <b>0.264</b>  | 36.83        | 8.76         | <b>13.53</b>    | <b>33.9</b>     | <b>0.001755</b>    |
| Phi-3-mini-4K                | Naive          | 7.532         | <b>7.14</b>  | <b>7.04</b>  | <b>0.00</b>     | <b>0.0</b>      | 0.001513           |
|                              | Hybrid         | 3.014         | 8.94         | 7.31         | <b>0.00</b>     | <b>0.0</b>      | <b>0.000676</b>    |
|                              | HAH            | <b>0.579</b>  | 12.01        | 7.80         | 55.74           | 139.4           | 0.004618           |
|                              | Reasoning      | 3.859         | 9.07         | 7.76         | <b>0.00</b>     | <b>0.0</b>      | 0.002062           |
|                              | Mini-Reasoning | 3.414         | 7.61         | 7.22         | <b>0.00</b>     | <b>0.0</b>      | 0.001352           |
| Llama-3-2-3B                 | Naive          | 2.186         | <b>12.30</b> | <b>6.76</b>  | 45.37           | 113.7           | 0.015226           |
|                              | Hybrid         | 1.421         | 15.68        | 7.37         | 72.11           | 179.3           | 0.014779           |
|                              | HAH            | 2.062         | 12.44        | 6.92         | 47.63           | 119.3           | 0.015173           |
|                              | Reasoning      | <b>0.532</b>  | 31.54        | 8.83         | 23.45           | 58.6            | <b>0.004874</b>    |
|                              | Mini-Reasoning | 0.652         | 34.25        | 8.92         | <b>22.95</b>    | <b>57.5</b>     | 0.005553           |

### 6.5.1 The Mechanics of Multi-Layer Retrieval

We begin with what we consider the most consequential architectural decision in HAH, the two-pass retrieval strategy. Understanding its impact requires first appreciating the limitations it addresses.

Single-pass retrieval, whether sparse or dense, operates under an implicit assumption, that the original query contains sufficient information to locate all relevant evidence. For simple factual queries, this assumption often holds. For multi-hop reasoning, it frequently does not. A question such as “What is the population of the country where the Eiffel Tower is located?” requires first identifying France as the relevant country before population statistics become discoverable. The original query contains no explicit mention of France, and a single retrieval pass may therefore return information about the Eiffel Tower without the demographic data needed to answer the question.

HAH’s two-pass mechanism addresses this limitation by treating initially retrieved content as a source of implicit query expansion. Having retrieved documents about the Eiffel Tower, the system can then issue secondary queries informed by this context, discovering information about France that the original query could not directly surface. This process, which we term “context-guided re-querying,” effectively enlarges the semantic footprint of the search.

The empirical consequences of this design prove substantial. We examined failure cases for Llama-3-8B on 2WikiMultiHop, where the Hybrid pipeline achieves factuality of only 0.222 compared to HAH’s 0.811. Manual inspection reveals a consistent pattern, Hybrid retrieval typically returns only partial evidence chains, capturing documents relevant to one reasoning hop whilst missing those needed for subsequent steps. HAH, through its secondary retrieval pass, captures what we might call “bridging documents”, content that connects disparate pieces of evidence into coherent reasoning chains. This nearly fourfold improvement in factuality reflects not merely better retrieval in degree, but retrieval that succeeds in kind where simpler approaches fail.

### 6.5.2 Contextual Compression as Attention Guidance

The contextual compression module, implemented through FlashReranker, serves purposes that extend beyond mere context window management. Whilst compression is indeed necessary for models with limited context capacity, we find evidence that its benefits are more fundamental.

Consider what compression entails at a conceptual level. The module must decide which retrieved content to retain and which to discard, implicitly encoding judgments about relevance. When functioning well, compression acts as a form of attention guidance, highlighting for the generative model the information most pertinent to the query at hand. When functioning poorly, it introduces information loss that manifests as degraded response quality.

Our experiments suggest that HAH’s compression succeeds in the former. Models with constrained context windows, most notably Phi-3-mini-4K with its four thousand token limit, achieve their strongest performance under HAH. This outcome would be difficult to explain if compression merely truncated context; it suggests instead that compression actively improves the signal-to-noise ratio of retrieved material.

We determined the compression ratio ( $r_c = 0.7$ ) through preliminary experiments, settling on a value that balances information retention against context efficiency. Lower compression ratios ( $r_c < 0.5$ ) introduced information loss visible in degraded correctness scores. Higher ratios ( $r_c > 0.85$ ) failed to provide meaningful filtering benefit. The chosen parameterisation represents a middle ground, though we note that adaptive compression, tuned to query or model characteristics, might yield further improvements.

### 6.5.3 Adapting to Query Complexity

Not all queries demand equal retrieval depth. A simple factual question requires fewer retrieved passages than a complex multi-hop reasoning task; retrieving excessively for simple queries risks introducing noise without corresponding benefit. HAH incorporates a query complexity analyser that dynamically adjusts retrieval parameters in response to estimated query difficulty.

The mechanism operates through surface-level query features, word count, syntactic complexity, and the presence of multiple sub-questions. Whilst admittedly coarse, these heuristics prove effective in practice. Simple queries receive shallower retrieval, reducing both latency and the risk of retrieving misleading context. Complex queries trigger deeper retrieval with adjusted fusion weights, ensuring sufficient evidential coverage.

We observe the consequences of this adaptation most clearly on datasets with heterogeneous query difficulty. CommonsenseQA, for instance, contains questions ranging from straightforward factual lookups to nuanced reasoning about everyday scenarios. The adaptive mechanism allows HAH to allocate retrieval resources in proportion to need, achieving efficiency on simple queries whilst maintaining thoroughness on complex ones. A fixed parameterisation would necessarily compromise on one objective or the other.

### 6.5.4 Understanding Reasoning Variant Underperformance

The Reasoning and Mini-Reasoning configurations, which augment HAH with Bayesian reasoning detection, exhibit notably lower performance on our benchmark suite. This finding warrants careful analysis, as it might superficially suggest that reasoning awareness is counterproductive.

We identify two contributing factors. The first concerns overhead. The reasoning detection mechanism, which classifies queries into categories such as factual, analytical, comparative, causal, and hypothetical, incurs non-trivial computational cost. For smaller models operating near resource limits, this overhead may not be offset by any improvement in prompt targeting. The classification itself is imperfect, and misclassification can direct queries to inappropriate prompt templates.

The second factor is more fundamental and concerns evaluation methodology. The reasoning-enhanced variants incorporate a “trivial question bypass” mechanism, designed to recognise simple conversational exchanges (greetings, acknowledgments, and the like) that require no retrieval. In conversational deployment, this mechanism would save unnecessary computation on exchanges such as **Hello** or **Thank you**. In benchmark evaluation, however, where each query is evaluated independently without conversational context, the mechanism misfires with troubling frequency.

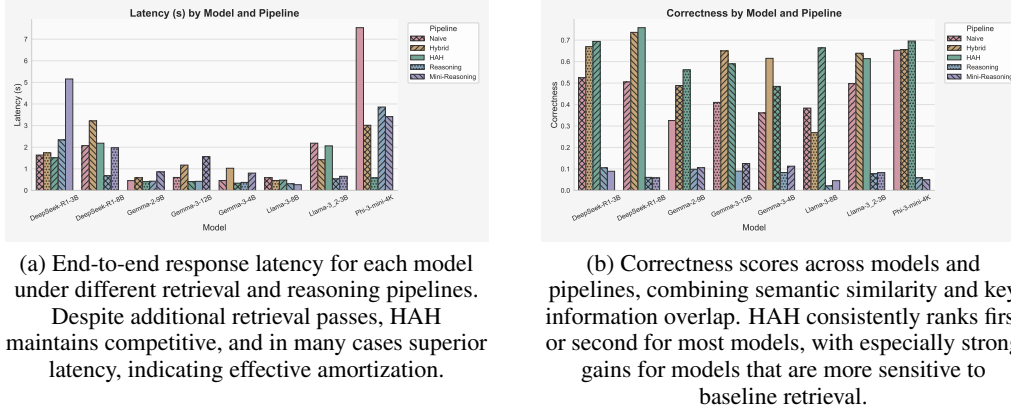


Figure 5: Latency and correctness trade-offs across models and pipeline configurations.

Queries that pattern-match to **trivial** criteria bypass the full retrieval pipeline, receiving responses generated without proper contextual grounding. The resulting answers, whilst fluent, lack the factual foundation that benchmark evaluation rewards. We observe this most clearly in cases where the Reasoning variant produces confident-sounding but incorrect responses, having bypassed the retrieval that would have supplied corrective information.

These observations do not, we believe, invalidate the reasoning-aware approach. They do, however, suggest that such mechanisms require deployment contexts where their assumptions hold, conversational applications with persistent context, where trivial exchanges are commonplace and where the overhead of reasoning detection can be amortised across extended interactions. Single-turn benchmark evaluation penalises these designs for virtues that would manifest as benefits in intended use cases.

### 6.5.5 Environmental and Economic Implications

Beyond task performance, our analysis reveals implications for sustainable deployment. The asynchronous architecture that enables HAH’s quality improvements also contributes to reduced resource consumption.

We observe that HAH achieves lower GPU utilisation than alternative pipelines across multiple model configurations. For several configurations, HAH maintains power consumption below 5W compared to over 100W observed for some Reasoning variants. The mechanism is straightforward, parallelised retrieval with intelligent caching eliminates redundant computation that sequential approaches must perform. Lower utilisation translates to reduced power draw and, consequently, to lower carbon emissions per query processed.

The aggregate impact, whilst small on a per-query basis, compounds at scale. We estimate CO<sub>2</sub> emission reductions of approximately 66% compared to traditional hybrid approaches. An organisation processing ten million queries daily could, through pipeline selection alone, reduce annual carbon emissions by several metric tons relative to a Hybrid deployment achieving inferior quality. This finding aligns with a broader trend in the field toward considering environmental cost as a first-class design criterion alongside task performance.

Economic considerations tell a similar story. HAH achieves cost parity or improvement relative to Naive retrieval for the majority of evaluated models, whilst delivering superior generation quality. The savings derive both from reduced computational overhead (lower per-query cost) and from improved quality (reduced need for costly human review or correction of generated content).

### 6.5.6 Model Architecture Interactions

We conclude our qualitative analysis by examining how model characteristics interact with pipeline effectiveness. Not all models respond equally to sophisticated retrieval; understanding these interactions informs practical deployment decisions.

We observe three distinct behavioural clusters. The first comprises instruction-following models, exemplified by Phi-3-mini-4K and DeepSeek-R1. These models, trained with explicit emphasis on following instructions and grounding responses in provided context, show the most consistent benefits from HAH. Phi-3-mini-4K achieves its peak factuality of 0.817 under HAH, suggesting that well-aligned models effectively leverage the higher-quality context provided by multi-layer retrieval. Their instruction-tuning appears to enhance their capacity to translate retrieval improvements directly into generation improvements.

The second cluster comprises base models such as Llama-3-8B and members of the Gemma family. These models exhibit more variable behaviour across pipelines, with some configurations producing markedly degraded outputs. Llama-3-8B under Hybrid retrieval produces relevance of only 0.314, well below the 0.764 achieved under HAH. We interpret this sensitivity as reflecting base models' greater reliance on explicit contextual grounding; when retrieval quality falters, they lack the parametric knowledge to compensate. HAH's more robust retrieval provides the stable foundation these models require.

The third cluster comprises distilled models, represented in our experiments by DeepSeek-R1-Distill variants. These models, derived from larger teachers through knowledge distillation, show HAH benefits primarily in efficiency rather than raw quality. DeepSeek-R1-Distill-Llama-8B achieves the highest overall correctness (0.758) and near-ceiling fluency (0.981) under HAH, but the absolute gains over Hybrid are more modest than for base models. The distillation process appears to have already optimised these models for efficient context utilisation. HAH nonetheless provides value through reduced latency and resource consumption, benefits that compound in deployment settings with strict efficiency requirements.

## 7 Conclusion

We set out to investigate whether the additional architectural complexity of multi-layer hybrid retrieval translates into meaningful improvements for retrieval-augmented generation. The evidence assembled in this paper suggests that it does, though the picture that emerges is more nuanced than a simple affirmative.

HAH-RAG, the framework we have presented, embodies several design principles that our experiments validate. The asynchronous execution of sparse and dense retrieval eliminates sequential bottlenecks that otherwise inflate latency without improving quality. The two-pass retrieval mechanism, wherein initially retrieved content informs subsequent search, expands semantic coverage in ways that prove consequential for multi-hop reasoning. Dynamic parameter adaptation allocates retrieval depth in proportion to query complexity, achieving efficiency where thoroughness is unnecessary and thoroughness where efficiency would compromise quality. Contextual compression with cross-encoder reranking filters noise whilst preserving the information most relevant to generation.

These mechanisms combine to yield improvements that, whilst varying across models and tasks, prove consistent in direction. Factuality gains are perhaps most notable; we observe improvements of approximately 6% on average, with particularly dramatic results on multi-hop reasoning tasks where HAH achieves nearly fourfold improvements over Hybrid baselines (0.811 versus 0.222 for Llama-3-8B on 2WikiMultiHop). The improvements prove most pronounced on tasks requiring evidence synthesis, where the two-pass mechanism demonstrates its intended function of capturing bridging evidence that single-pass retrieval misses. On simpler extractive tasks, HAH maintains competitive performance without introducing prohibitive overhead.

The efficiency characteristics of HAH merit emphasis. Despite its architectural complexity, HAH achieves latency competitive with or superior to baseline configurations. GPU utilisation and power consumption decrease rather than increase, with estimated CO<sub>2</sub> emission reductions of approximately 66% compared to traditional hybrid approaches. These environmental implications, whilst modest on a per-query basis, compound meaningfully at the scale of production deployment.

We acknowledge, however, that our reasoning-enhanced variants, which augment HAH with Bayesian reasoning detection and trivial question bypass, underperform on the benchmark evaluation we conducted. This finding reflects not a fundamental flaw in reasoning awareness, but rather a mismatch between mechanism and evaluation context. Single-turn benchmark evaluation penalises designs optimised for conversational deployment, where trivial exchanges are common and where

Table 7: Resource Usage and Cost Analysis, Latency (s), GPU Utilization (%), and Total Cost (€) across models, pipelines, and datasets. Best (lowest) values per model-dataset combination are shown in **bold**.

| Model                 | Pipeline       | SQuAD           | CommonsenseQA   | HotpotQA        | 2WikiMultiHop   | AmbigQA         | Avg      |
|-----------------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------|
| <i>Latency (s)</i>    |                |                 |                 |                 |                 |                 |          |
| DeepSeek-R1-3B        | Naive          | 1.434           | 1.901           | 1.635           | 1.588           | 1.622           | 1.636    |
|                       | Hybrid         | 1.652           | 2.283           | 1.639           | 1.372           | 1.726           | 1.735    |
|                       | HAH            | 1.735           | 1.401           | 1.419           | 1.184           | 1.807           | 1.509    |
|                       | Reasoning      | 0.379           | 0.332           | 0.468           | 0.367           | 0.265           | 0.362    |
|                       | Mini-Reasoning | <b>0.230</b>    | <b>0.105</b>    | <b>0.264</b>    | <b>0.301</b>    | <b>0.101</b>    | 0.200    |
| Gemma-2-9B            | Naive          | 0.387           | 0.510           | 0.423           | 0.419           | 0.531           | 0.454    |
|                       | Hybrid         | 0.574           | 0.866           | 0.472           | 0.390           | 0.619           | 0.584    |
|                       | HAH            | 0.393           | 0.379           | 0.424           | 0.436           | 0.395           | 0.406    |
|                       | Reasoning      | 0.394           | 0.348           | 0.485           | 0.360           | 0.277           | 0.373    |
|                       | Mini-Reasoning | <b>0.245</b>    | <b>0.115</b>    | <b>0.276</b>    | <b>0.324</b>    | <b>0.118</b>    | 0.215    |
| Gemma-3-4B            | Naive          | 0.431           | 0.424           | 0.366           | 0.383           | 0.685           | 0.458    |
|                       | Hybrid         | 1.132           | 1.614           | 0.780           | 0.639           | 0.961           | 1.025    |
|                       | HAH            | 0.328           | 0.328           | 0.344           | 0.349           | 0.328           | 0.335    |
|                       | Reasoning      | 0.382           | 0.341           | 0.492           | 0.372           | 0.269           | 0.371    |
|                       | Mini-Reasoning | <b>0.246</b>    | <b>0.116</b>    | <b>0.291</b>    | <b>0.330</b>    | <b>0.117</b>    | 0.220    |
| Llama-3-8B            | Naive          | 0.457           | 0.673           | 0.531           | 0.445           | 0.839           | 0.589    |
|                       | Hybrid         | <b>0.391</b>    | 0.410           | 0.435           | <b>0.331</b>    | 0.728           | 0.459    |
|                       | HAH            | 0.466           | 0.484           | <b>0.432</b>    | 0.499           | <b>0.516</b>    | 0.479    |
|                       | Reasoning      | 0.600           | 0.532           | 0.684           | 0.543           | 0.625           | 0.597    |
|                       | Mini-Reasoning | 0.562           | <b>0.376</b>    | 0.623           | 0.593           | 0.588           | 0.548    |
| Phi-3-mini-4K         | Naive          | 7.305           | 6.639           | 7.762           | 7.870           | 9.366           | 7.788    |
|                       | Reasoning      | <b>0.389</b>    | <b>0.351</b>    | <b>0.552</b>    | <b>0.455</b>    | <b>0.325</b>    | 0.414    |
| SmolLM3-3B            | Reasoning      | 0.499           | 0.425           | 0.562           | 0.374           | 0.269           | 0.426    |
|                       | Mini-Reasoning | <b>0.253</b>    | <b>0.114</b>    | <b>0.273</b>    | <b>0.319</b>    | <b>0.103</b>    | 0.212    |
| <i>GPU Util. (%)</i>  |                |                 |                 |                 |                 |                 |          |
| DeepSeek-R1-3B        | Naive          | 0.48            | –               | 1.96            | 2.98            | 3.55            | 1.79     |
|                       | Hybrid         | 0.43            | <b>0.22</b>     | <b>1.94</b>     | 3.59            | 3.82            | 2.00     |
|                       | HAH            | <b>0.41</b>     | –               | 2.02            | <b>2.56</b>     | 3.51            | 1.70     |
|                       | Reasoning      | 4.54            | 4.06            | 4.69            | 3.68            | <b>3.44</b>     | 4.08     |
|                       | Mini-Reasoning | 3.03            | 3.90            | 1.98            | 2.84            | 4.39            | 3.23     |
| Gemma-2-9B            | Naive          | 1.12            | 1.60            | 1.46            | <b>0.94</b>     | 1.70            | 1.36     |
|                       | Hybrid         | <b>0.92</b>     | 1.88            | <b>0.74</b>     | 1.32            | 1.42            | 1.26     |
|                       | HAH            | 1.15            | 1.29            | 1.50            | 1.08            | 1.21            | 1.24     |
|                       | Reasoning      | 1.46            | 1.25            | 1.64            | 1.14            | 0.43            | 1.18     |
|                       | Mini-Reasoning | 2.13            | <b>0.21</b>     | 1.40            | 1.43            | <b>0.11</b>     | 1.05     |
| Gemma-3-4B            | Naive          | 1.47            | 0.15            | 1.56            | 1.21            | 1.25            | 1.13     |
|                       | Hybrid         | 1.37            | 1.16            | <b>1.48</b>     | 1.21            | 1.59            | 1.36     |
|                       | HAH            | 1.86            | 0.17            | 1.60            | 1.23            | <b>0.23</b>     | 1.02     |
|                       | Reasoning      | <b>1.26</b>     | 1.12            | 1.95            | <b>1.01</b>     | 0.46            | 1.16     |
|                       | Mini-Reasoning | 1.68            | <b>0.08</b>     | 1.95            | 1.33            | 0.32            | 1.07     |
| Llama-3-8B            | Naive          | 50.85           | 61.46           | 55.13           | 49.07           | 63.32           | 55.97    |
|                       | Hybrid         | 48.10           | 58.91           | 59.27           | 52.73           | 48.76           | 53.55    |
|                       | HAH            | 49.67           | 61.79           | 55.60           | 51.16           | 59.74           | 55.59    |
|                       | Reasoning      | <b>1.99</b>     | 1.85            | <b>1.45</b>     | <b>1.70</b>     | <b>0.82</b>     | 1.56     |
|                       | Mini-Reasoning | 2.15            | <b>0.39</b>     | 1.49            | 1.84            | 1.17            | 1.41     |
| Phi-3-mini-4K         | Naive          | 15.52           | <b>0.99</b>     | <b>0.12</b>     | <b>0.86</b>     | <b>1.20</b>     | 3.74     |
|                       | Reasoning      | <b>4.34</b>     | 3.69            | 3.67            | 2.90            | 2.93            | 3.50     |
| SmolLM3-3B            | Reasoning      | 3.45            | <b>2.84</b>     | 2.48            | <b>1.10</b>     | <b>2.83</b>     | 2.54     |
|                       | Mini-Reasoning | <b>2.65</b>     | 3.32            | <b>1.78</b>     | 2.69            | 4.28            | 2.94     |
| <i>Total Cost (€)</i> |                |                 |                 |                 |                 |                 |          |
| DeepSeek-R1-3B        | Naive          | <b>0.000655</b> | <b>0.000402</b> | 0.001463        | 0.003731        | 0.001908        | 0.001632 |
|                       | Hybrid         | <b>0.000723</b> | <b>0.000548</b> | 0.001562        | 0.003742        | 0.002360        | 0.001787 |
|                       | HAH            | <b>0.000717</b> | <b>0.000306</b> | 0.001298        | 0.002324        | 0.002125        | 0.001354 |
|                       | Reasoning      | <b>0.000842</b> | <b>0.000346</b> | <b>0.000941</b> | <b>0.001115</b> | <b>0.000243</b> | 0.000697 |
|                       | Mini-Reasoning | <b>0.000314</b> | <b>0.000156</b> | <b>0.000272</b> | <b>0.000551</b> | <b>0.000139</b> | 0.000286 |
| Gemma-2-9B            | Naive          | <b>0.000179</b> | <b>0.000149</b> | <b>0.000175</b> | <b>0.000231</b> | <b>0.000147</b> | 0.000176 |
|                       | Hybrid         | <b>0.000264</b> | <b>0.000279</b> | <b>0.000145</b> | <b>0.000223</b> | <b>0.000188</b> | 0.000220 |
|                       | HAH            | <b>0.000187</b> | <b>0.000094</b> | <b>0.000174</b> | <b>0.000262</b> | <b>0.000095</b> | 0.000162 |
|                       | Reasoning      | <b>0.000867</b> | <b>0.000334</b> | <b>0.000862</b> | <b>0.000740</b> | <b>0.000121</b> | 0.000585 |
|                       | Mini-Reasoning | <b>0.000459</b> | <b>0.000163</b> | <b>0.000448</b> | <b>0.000920</b> | <b>0.000183</b> | 0.000435 |
| Gemma-3-4B            | Naive          | <b>0.000230</b> | <b>0.000063</b> | <b>0.000154</b> | <b>0.000231</b> | <b>0.000324</b> | 0.000200 |
|                       | Hybrid         | <b>0.000601</b> | <b>0.000798</b> | 0.001480        | 0.001374        | <b>0.000197</b> | 0.000890 |
|                       | HAH            | <b>0.000204</b> | <b>0.000048</b> | <b>0.000140</b> | <b>0.000222</b> | <b>0.000046</b> | 0.000132 |
|                       | Reasoning      | <b>0.000631</b> | <b>0.000411</b> | <b>0.000950</b> | <b>0.001034</b> | <b>0.000134</b> | 0.000632 |
|                       | Mini-Reasoning | <b>0.000509</b> | <b>0.000184</b> | <b>0.000564</b> | <b>0.001029</b> | <b>0.000188</b> | 0.000495 |
| Llama-3-8B            | Naive          | 0.004254        | 0.003716        | 0.004046        | 0.005025        | 0.005021        | 0.004412 |
|                       | Hybrid         | 0.003422        | 0.002212        | 0.003483        | 0.003981        | 0.003148        | 0.003249 |
|                       | HAH            | 0.004228        | 0.002699        | 0.003316        | 0.005714        | 0.002779        | 0.003747 |
|                       | Reasoning      | <b>0.000848</b> | <b>0.000361</b> | <b>0.000868</b> | <b>0.001264</b> | <b>0.000243</b> | 0.000717 |
|                       | Mini-Reasoning | <b>0.000731</b> | <b>0.000281</b> | <b>0.000739</b> | <b>0.001388</b> | <b>0.000447</b> | 0.000717 |
| Phi-3-mini-4K         | Naive          | 0.020683        | <b>0.000964</b> | <b>0.000834</b> | 0.003195        | 0.002072        | 0.005550 |
|                       | Reasoning      | <b>0.001034</b> | <b>0.000529</b> | <b>0.001550</b> | <b>0.002167</b> | <b>0.000587</b> | 0.001173 |
| SmolLM3-3B            | Reasoning      | 0.001913        | <b>0.000825</b> | 0.001465        | <b>0.001074</b> | <b>0.000398</b> | 0.001133 |
|                       | Mini-Reasoning | <b>0.000639</b> | <b>0.000231</b> | <b>0.000412</b> | <b>0.000641</b> | <b>0.000158</b> | 0.000416 |

reasoning overhead can be amortised across extended interactions. The appropriate lesson is not that reasoning awareness lacks value, but that deployment context must inform architectural decisions.

## 7.1 Limitations and Future Directions

We conclude by acknowledging limitations that bound the claims we can responsibly make and by identifying directions that future research might productively pursue.

Our evaluation focuses exclusively on English-language benchmarks. Whilst we have no reason to expect HAH’s core mechanisms to fail in other languages, the BM25 component relies on lexical matching that may require language-specific tuning. Morphologically rich languages, in particular, may demand stemming or lemmatisation strategies that our current implementation does not provide. Cross-lingual retrieval, wherein queries and documents appear in different languages, presents additional challenges that we have not addressed.

The hyperparameters governing HAH, including compression ratios, fusion weights, and complexity thresholds, were determined through limited empirical search. We selected values that performed well across our evaluation suite, but we cannot claim optimality. A more principled approach might learn these parameters from data, perhaps through meta-learning across datasets or through end-to-end optimisation that propagates generation quality signals back to retrieval configuration. The fixed thresholds for query complexity classification, in particular, may not generalise optimally to domains with linguistic characteristics differing from those of our benchmarks.

Our evaluation encompasses models ranging from three to twelve billion parameters. The behaviour of HAH with substantially larger models remains uncharacterised. We speculate that as model capacity increases and parametric knowledge becomes more comprehensive, the relative benefits of sophisticated retrieval may diminish. Alternatively, larger models may prove more capable of leveraging high-quality retrieved context, amplifying rather than diminishing HAH’s advantages. Empirical investigation is needed to resolve this question.

The benchmark paradigm we employ, treating each query independently without conversational history, may not reflect deployment patterns where extended interactions are the norm. Our reasoning-enhanced variants, which underperform in benchmark evaluation, might excel in conversational settings for which they were designed. Future work should evaluate HAH in dialogue contexts where its adaptive mechanisms encounter the conditions they anticipate.

Several promising directions suggest themselves for future research. Learned parameter adaptation, replacing rule-based complexity analysis with models trained to optimise retrieval configuration, could improve both performance and generalisation. Integration with instruction-tuned retrievers, capable of interpreting natural language retrieval specifications, might enable more sophisticated query understanding. Extension to multimodal retrieval, incorporating images and structured data alongside text, would broaden HAH’s applicability. Investigation in domain-specific applications, particularly legal, medical, and scientific contexts where factuality carries heightened importance, could demonstrate practical value and surface domain-specific challenges.

Finally, the intersection of efficient retrieval architectures with emerging long-context models presents questions we have not addressed. As context windows expand toward millions of tokens, the calculus underlying retrieval-augmented generation shifts. When an entire document collection fits within context, what role remains for retrieval? We suspect the answer is **a substantial one**, since attention over millions of tokens scales poorly, and retrieval can provide the focusing function that raw context inclusion cannot. But this remains speculation; the empirical work needed to evaluate it lies ahead.

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## A Evaluation Dataset Sources

### 1. Knowledge-intensive QA:

- Natural Questions (NQ): <https://github.com/google-research-datasets/natural-questions>
- TriviaQA: <http://nlp.cs.washington.edu/triviaqa/>

### 2. Multi-hop Reasoning:

- HotpotQA: <https://hotpotqa.github.io/>

### 3. Long-form Generation:

- NarrativeQA: <https://github.com/deepmind/narrativeqa>

Most datasets are provided in JSON format, and some offer additional formats such as CSV or SQL databases. Although each data set includes specific splits for training, validation, and testing purposes; we only use the data set evaluation in the data set evaluation for performance comparison.

## B Further Quantitative Evaluation



Figure 6: Evolution of quality metrics over evaluation instances on all datasets for a fixed dataset length.

Table 8: Dataset-specific performance by model and pipeline. Up arrows indicate higher values are preferred; best results per dataset are shown in **bold**. The table highlights that gains from advanced pipelines are dataset-dependent, with multi-hop datasets exhibiting larger improvements in coherence and relevance.

| Model                        | Pipeline       | NDCG ↑       | R-1 ↑        | R-2 ↑        | R-L ↑        | Fluency ↑    | Coherence ↑  | Relevance ↑  | Factuality ↑ | Correctness ↑ |
|------------------------------|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|
| <b>SQuAD</b>                 |                |              |              |              |              |              |              |              |              |               |
| DeepSeek-R1-3B               | Naive          | <b>0.863</b> | 0.130        | 0.065        | 0.130        | 0.858        | 0.255        | 0.716        | 0.737        | 0.547         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.806        | 0.330        | 0.716        | 0.717        | 0.637         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.824        | 0.335        | 0.751        | 0.753        | 0.654         |
|                              | Reasoning      | 0.417        | 0.016        | 0.007        | 0.016        | 0.354        | 0.141        | 0.325        | 0.246        | 0.151         |
|                              | Mini-Reasoning | 0.437        | 0.024        | 0.013        | 0.024        | 0.359        | 0.155        | 0.323        | 0.250        | 0.156         |
| DeepSeek-R1-Distill-Llama-8B | Naive          | <b>0.863</b> | 0.033        | 0.012        | 0.033        | 0.802        | 0.358        | 0.723        | 0.709        | 0.501         |
|                              | Hybrid         | <b>0.863</b> | 0.023        | 0.000        | 0.023        | 0.890        | 0.382        | 0.744        | 0.745        | 0.723         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | <b>0.959</b> | 0.352        | 0.708        | 0.709        | <b>0.754</b>  |
|                              | Reasoning      | 0.417        | 0.003        | 0.001        | 0.003        | 0.144        | 0.060        | 0.145        | 0.102        | 0.072         |
|                              | Mini-Reasoning | 0.437        | 0.006        | 0.002        | 0.005        | 0.158        | 0.061        | 0.159        | 0.111        | 0.078         |
| Gemma-2-9B                   | Naive          | <b>0.863</b> | 0.111        | 0.000        | 0.111        | 0.589        | 0.083        | 0.628        | 0.571        | 0.332         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.501        | 0.147        | 0.551        | 0.535        | 0.440         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.743        | 0.057        | 0.694        | 0.688        | 0.605         |
|                              | Reasoning      | 0.419        | 0.062        | 0.039        | 0.061        | 0.359        | 0.040        | 0.342        | 0.230        | 0.126         |
|                              | Mini-Reasoning | 0.437        | 0.041        | 0.024        | 0.040        | 0.397        | 0.075        | 0.338        | 0.257        | 0.155         |
| Gemma-3-12B                  | Naive          | <b>0.863</b> | <b>0.353</b> | <b>0.267</b> | <b>0.353</b> | 0.783        | 0.130        | 0.761        | 0.695        | 0.424         |
|                              | Hybrid         | <b>0.863</b> | 0.118        | 0.000        | 0.118        | 0.768        | 0.299        | 0.732        | 0.732        | 0.625         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.736        | 0.060        | 0.662        | 0.664        | 0.594         |
|                              | Reasoning      | 0.415        | 0.089        | 0.057        | 0.089        | 0.327        | 0.032        | 0.293        | 0.199        | 0.109         |
|                              | Mini-Reasoning | 0.437        | 0.038        | 0.024        | 0.037        | 0.385        | 0.071        | 0.330        | 0.258        | 0.156         |
| Gemma-3-4B                   | Naive          | <b>0.863</b> | 0.316        | 0.235        | 0.316        | 0.617        | 0.125        | 0.632        | 0.589        | 0.356         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.722        | 0.311        | 0.718        | 0.719        | 0.602         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.627        | 0.061        | 0.636        | 0.640        | 0.525         |
|                              | Reasoning      | 0.420        | 0.123        | 0.075        | 0.123        | 0.279        | 0.037        | 0.297        | 0.189        | 0.102         |
|                              | Mini-Reasoning | 0.437        | 0.033        | 0.019        | 0.032        | 0.396        | 0.098        | 0.332        | 0.261        | 0.161         |
| Llama-3-8B                   | Naive          | <b>0.863</b> | 0.316        | 0.235        | 0.316        | 0.698        | 0.163        | 0.676        | 0.625        | 0.385         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.228        | 0.089        | 0.259        | 0.198        | 0.207         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.796        | 0.152        | 0.776        | 0.774        | 0.658         |
|                              | Reasoning      | 0.417        | 0.010        | 0.006        | 0.010        | 0.061        | 0.018        | 0.096        | 0.039        | 0.031         |
|                              | Mini-Reasoning | 0.437        | 0.014        | 0.008        | 0.014        | 0.093        | 0.021        | 0.115        | 0.071        | 0.048         |
| Phi-3-mini-4K                | Naive          | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.844        | <b>0.409</b> | 0.665        | 0.755        | 0.648         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.784        | 0.326        | 0.707        | 0.708        | 0.628         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.846        | 0.103        | <b>0.816</b> | <b>0.818</b> | 0.696         |
|                              | Reasoning      | 0.414        | 0.016        | 0.011        | 0.016        | 0.148        | 0.104        | 0.226        | 0.178        | 0.092         |
|                              | Mini-Reasoning | 0.437        | 0.005        | 0.002        | 0.005        | 0.161        | 0.101        | 0.213        | 0.160        | 0.082         |
| DeepSeek-R1-Distill-Llama-3B | Naive          | <b>0.863</b> | 0.118        | 0.070        | 0.118        | 0.834        | 0.397        | 0.695        | 0.693        | 0.493         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.727        | 0.375        | 0.693        | 0.694        | 0.597         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.719        | 0.380        | 0.704        | 0.705        | 0.592         |
|                              | Reasoning      | 0.415        | 0.027        | 0.016        | 0.026        | 0.301        | 0.101        | 0.249        | 0.186        | 0.110         |
|                              | Mini-Reasoning | 0.437        | 0.015        | 0.008        | 0.015        | 0.251        | 0.115        | 0.202        | 0.184        | 0.122         |
| <b>CommonsenseQA</b>         |                |              |              |              |              |              |              |              |              |               |
| Model                        | Pipeline       | NDCG ↑       | R-1 ↑        | R-2 ↑        | R-L ↑        | Fluency ↑    | Coherence ↑  | Relevance ↑  | Factuality ↑ | Correctness ↑ |
| DeepSeek-R1-3B               | Naive          | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.899        | 0.301        | 0.676        | 0.720        | 0.532         |
|                              | Hybrid         | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.855        | 0.309        | 0.678        | 0.677        | 0.637         |
|                              | HAH            | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.884        | 0.362        | 0.743        | 0.746        | 0.697         |
|                              | Reasoning      | 0.455        | 0.003        | 0.000        | 0.003        | 0.402        | 0.140        | 0.306        | 0.105        | 0.078         |
|                              | Mini-Reasoning | 0.461        | 0.005        | 0.001        | 0.005        | 0.384        | 0.155        | 0.327        | 0.056        | 0.044         |
| DeepSeek-R1-Distill-Llama-8B | Naive          | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.773        | 0.407        | 0.690        | 0.702        | 0.489         |
|                              | Hybrid         | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.891        | 0.379        | 0.708        | 0.706        | 0.695         |
|                              | HAH            | <b>0.974</b> | 0.000        | 0.000        | 0.000        | <b>0.980</b> | 0.355        | 0.672        | 0.676        | <b>0.757</b>  |
|                              | Reasoning      | 0.454        | 0.001        | 0.000        | 0.001        | 0.155        | 0.056        | 0.129        | 0.092        | 0.072         |
|                              | Mini-Reasoning | 0.461        | 0.005        | 0.002        | 0.005        | 0.360        | 0.164        | 0.250        | 0.088        | 0.061         |
| Gemma-2-9B                   | Naive          | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.566        | 0.167        | 0.643        | 0.604        | 0.346         |
|                              | Hybrid         | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.589        | 0.262        | 0.649        | 0.640        | 0.508         |
|                              | HAH            | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.693        | 0.063        | 0.706        | 0.709        | 0.583         |
|                              | Reasoning      | 0.454        | 0.033        | 0.008        | 0.033        | 0.288        | 0.077        | 0.313        | 0.137        | 0.076         |
|                              | Mini-Reasoning | 0.461        | 0.009        | 0.003        | 0.009        | 0.412        | 0.112        | 0.325        | 0.079        | 0.057         |
| Gemma-3-12B                  | Naive          | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.776        | 0.181        | 0.712        | 0.678        | 0.409         |
|                              | Hybrid         | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.746        | 0.331        | 0.706        | 0.705        | 0.602         |
|                              | HAH            | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.832        | 0.073        | 0.726        | 0.730        | 0.636         |
|                              | Reasoning      | 0.455        | <b>0.060</b> | <b>0.018</b> | <b>0.060</b> | 0.289        | 0.053        | 0.275        | 0.148        | 0.082         |
|                              | Mini-Reasoning | 0.461        | 0.007        | 0.003        | 0.007        | 0.414        | 0.083        | 0.265        | 0.189        | 0.123         |
| Gemma-3-4B                   | Naive          | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.495        | 0.154        | 0.585        | 0.578        | 0.337         |
|                              | Hybrid         | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.677        | 0.320        | 0.687        | 0.686        | 0.565         |
|                              | HAH            | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.608        | 0.082        | 0.603        | 0.611        | 0.469         |
|                              | Reasoning      | 0.455        | 0.058        | 0.013        | 0.058        | 0.227        | 0.052        | 0.283        | 0.138        | 0.076         |
|                              | Mini-Reasoning | 0.461        | 0.008        | 0.003        | 0.008        | 0.459        | 0.099        | 0.304        | 0.146        | 0.102         |
| Llama-3-8B                   | Naive          | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.715        | 0.261        | 0.649        | 0.624        | 0.391         |
|                              | Hybrid         | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.196        | 0.078        | 0.224        | 0.164        | 0.179         |
|                              | HAH            | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.857        | 0.191        | 0.761        | 0.759        | 0.672         |
|                              | Reasoning      | 0.455        | 0.000        | 0.000        | 0.000        | 0.009        | 0.004        | 0.047        | 0.006        | 0.012         |
|                              | Mini-Reasoning | 0.461        | 0.001        | 0.000        | 0.001        | 0.031        | 0.013        | 0.064        | 0.021        | 0.032         |
| Phi-3-mini-4K                | Naive          | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.932        | 0.343        | 0.657        | <b>0.770</b> | 0.665         |
|                              | Hybrid         | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.838        | 0.292        | 0.705        | 0.703        | 0.635         |
|                              | HAH            | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.773        | 0.109        | <b>0.769</b> | 0.769        | 0.613         |
|                              | Reasoning      | 0.455        | 0.013        | 0.002        | 0.013        | 0.079        | 0.115        | 0.227        | 0.085        | 0.045         |
|                              | Mini-Reasoning | 0.461        | 0.000        | 0.000        | 0.000        | 0.093        | 0.115        | 0.221        | 0.073        | 0.038         |
| DeepSeek-R1-Distill-Llama-3B | Naive          | <b>0.974</b> | 0.000        | 0.000        | 0.000        | 0.868        | <b>0.429</b> | 0.657        | 0.702        | 0.510         |

Continued on next page

| Model | Pipeline       | NDCG ↑       | R-1 ↑ | R-2 ↑ | R-L ↑ | Fluency ↑ | Coherence ↑ | Relevance ↑ | Factuality ↑ | Correctness ↑ |
|-------|----------------|--------------|-------|-------|-------|-----------|-------------|-------------|--------------|---------------|
|       | Hybrid         | <b>0.974</b> | 0.000 | 0.000 | 0.000 | 0.782     | 0.369       | 0.653       | 0.650        | 0.586         |
|       | HAH            | <b>0.974</b> | 0.000 | 0.000 | 0.000 | 0.738     | 0.337       | 0.681       | 0.682        | 0.591         |
|       | Reasoning      | 0.455        | 0.002 | 0.000 | 0.002 | 0.255     | 0.107       | 0.219       | 0.083        | 0.057         |
|       | Mini-Reasoning | 0.461        | 0.002 | 0.001 | 0.002 | 0.201     | 0.094       | 0.166       | 0.078        | 0.058         |

#### HotpotQA

| Model                        | Pipeline       | NDCG ↑       | R-1 ↑        | R-2 ↑        | R-L ↑        | Fluency ↑    | Coherence ↑  | Relevance ↑  | Factuality ↑ | Correctness ↑ |
|------------------------------|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|
| DeepSeek-R1-3B               | Naive          | <b>0.863</b> | 0.064        | 0.015        | 0.064        | 0.947        | 0.220        | 0.762        | 0.756        | 0.541         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.953        | 0.302        | 0.758        | 0.758        | 0.709         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.954        | 0.302        | 0.801        | 0.808        | 0.723         |
|                              | Reasoning      | 0.422        | 0.016        | 0.007        | 0.016        | 0.417        | 0.130        | 0.325        | 0.124        | 0.093         |
|                              | Mini-Reasoning | 0.426        | 0.026        | 0.011        | 0.025        | 0.416        | 0.139        | 0.320        | 0.106        | 0.081         |
| DeepSeek-R1-Distill-Llama-8B | Naive          | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.936        | 0.350        | 0.762        | 0.730        | 0.525         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.979        | 0.353        | 0.774        | 0.774        | <b>0.763</b>  |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | <b>0.989</b> | 0.335        | 0.731        | 0.740        | 0.756         |
|                              | Reasoning      | 0.421        | 0.003        | 0.001        | 0.003        | 0.163        | 0.054        | 0.144        | 0.086        | 0.057         |
|                              | Mini-Reasoning | 0.426        | 0.005        | 0.002        | 0.005        | 0.181        | 0.052        | 0.152        | 0.073        | 0.053         |
| Gemma-2-9B                   | Naive          | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.611        | 0.067        | 0.551        | 0.519        | 0.295         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.668        | 0.097        | 0.607        | 0.599        | 0.501         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.713        | 0.048        | 0.640        | 0.656        | 0.527         |
|                              | Reasoning      | 0.420        | 0.054        | 0.035        | 0.054        | 0.329        | 0.035        | 0.284        | 0.159        | 0.087         |
|                              | Mini-Reasoning | 0.426        | 0.030        | 0.015        | 0.030        | 0.423        | 0.076        | 0.329        | 0.130        | 0.088         |
| Gemma-3-12B                  | Naive          | <b>0.863</b> | <b>0.105</b> | 0.000        | <b>0.105</b> | 0.879        | 0.085        | 0.764        | 0.709        | 0.417         |
|                              | Hybrid         | <b>0.863</b> | 0.091        | 0.000        | 0.091        | 0.944        | 0.242        | 0.771        | 0.772        | 0.690         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.791        | 0.052        | 0.679        | 0.692        | 0.586         |
|                              | Reasoning      | 0.421        | 0.058        | 0.036        | 0.057        | 0.339        | 0.032        | 0.275        | 0.157        | 0.086         |
|                              | Mini-Reasoning | 0.426        | 0.032        | 0.017        | 0.032        | 0.394        | 0.066        | 0.294        | 0.166        | 0.113         |
| Gemma-3-4B                   | Naive          | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.709        | 0.097        | 0.596        | 0.579        | 0.333         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.876        | 0.245        | 0.741        | 0.741        | 0.617         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.496        | 0.052        | 0.452        | 0.474        | 0.358         |
|                              | Reasoning      | 0.423        | 0.066        | <b>0.040</b> | 0.065        | 0.257        | 0.025        | 0.245        | 0.131        | 0.070         |
|                              | Mini-Reasoning | 0.426        | 0.031        | <b>0.015</b> | 0.030        | 0.431        | 0.071        | 0.316        | 0.132        | 0.094         |
| Llama-3-8B                   | Naive          | <b>0.863</b> | 0.037        | 0.000        | 0.037        | 0.776        | 0.144        | 0.663        | 0.624        | 0.376         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.417        | 0.097        | 0.359        | 0.332        | 0.318         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.771        | 0.113        | 0.684        | 0.679        | 0.596         |
|                              | Reasoning      | 0.421        | 0.004        | 0.002        | 0.004        | 0.058        | 0.009        | 0.071        | 0.024        | 0.018         |
|                              | Mini-Reasoning | 0.426        | 0.015        | 0.007        | 0.015        | 0.134        | 0.019        | 0.127        | 0.079        | 0.051         |
| Phi-3-mini-4K                | Naive          | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.939        | 0.303        | 0.657        | 0.792        | 0.672         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.940        | 0.252        | 0.761        | 0.761        | 0.700         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.931        | 0.094        | <b>0.833</b> | <b>0.837</b> | 0.692         |
|                              | Reasoning      | 0.422        | 0.007        | 0.003        | 0.007        | 0.178        | 0.104        | 0.214        | 0.099        | 0.052         |
|                              | Mini-Reasoning | 0.426        | 0.003        | 0.001        | 0.003        | 0.172        | 0.108        | 0.197        | 0.075        | 0.040         |
| DeepSeek-R1-Distill-Llama-3B | Naive          | <b>0.863</b> | 0.008        | 0.000        | 0.008        | 0.929        | <b>0.405</b> | 0.729        | 0.723        | 0.507         |
|                              | Hybrid         | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.910        | 0.344        | 0.756        | 0.755        | 0.678         |
|                              | HAH            | <b>0.863</b> | 0.000        | 0.000        | 0.000        | 0.870        | 0.385        | 0.721        | 0.727        | 0.618         |
|                              | Reasoning      | 0.421        | 0.013        | 0.005        | 0.013        | 0.358        | 0.115        | 0.272        | 0.123        | 0.081         |
|                              | Mini-Reasoning | 0.426        | 0.013        | 0.006        | 0.013        | 0.303        | 0.117        | 0.197        | 0.123        | 0.092         |

#### 2WikiMultiHop

| Model                        | Pipeline       | NDCG         | R-1          | R-2          | R-L          | Fluency      | Coherence | Relevance    | Factuality   | Correctness  |
|------------------------------|----------------|--------------|--------------|--------------|--------------|--------------|-----------|--------------|--------------|--------------|
| DeepSeek-R1-3B               | Naive          | <b>0.937</b> | 0.067        | 0.000        | 0.067        | 0.900        | 0.260     | 0.772        | 0.775        | 0.553        |
|                              | Hybrid         | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.945        | 0.306     | 0.783        | 0.782        | 0.709        |
|                              | HAH            | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.957        | 0.349     | 0.822        | 0.822        | 0.720        |
|                              | Reasoning      | 0.394        | 0.015        | 0.007        | 0.015        | 0.368        | 0.140     | 0.308        | 0.181        | 0.123        |
|                              | Mini-Reasoning | 0.410        | 0.033        | 0.016        | 0.032        | 0.374        | 0.157     | 0.308        | 0.172        | 0.116        |
| DeepSeek-R1-Distill-Llama-8B | Naive          | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.909        | 0.343     | 0.767        | 0.747        | 0.553        |
|                              | Hybrid         | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.987        | 0.351     | 0.767        | 0.766        | <b>0.748</b> |
|                              | HAH            | <b>0.937</b> | 0.000        | 0.000        | 0.000        | <b>0.996</b> | 0.317     | 0.697        | 0.697        | 0.746        |
|                              | Reasoning      | 0.394        | 0.003        | 0.001        | 0.003        | 0.154        | 0.053     | 0.131        | 0.104        | 0.063        |
|                              | Mini-Reasoning | 0.410        | 0.004        | 0.002        | 0.004        | 0.133        | 0.043     | 0.114        | 0.086        | 0.053        |
| Gemma-2-9B                   | Naive          | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.549        | 0.065     | 0.512        | 0.493        | 0.279        |
|                              | Hybrid         | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.474        | 0.074     | 0.462        | 0.453        | 0.363        |
|                              | HAH            | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.550        | 0.020     | 0.628        | 0.627        | 0.487        |
|                              | Reasoning      | 0.395        | 0.038        | 0.027        | 0.038        | 0.378        | 0.030     | 0.331        | 0.207        | 0.116        |
|                              | Mini-Reasoning | 0.410        | 0.038        | 0.023        | 0.037        | 0.395        | 0.057     | 0.334        | 0.196        | 0.122        |
| Gemma-3-12B                  | Naive          | <b>0.937</b> | <b>0.700</b> | <b>0.667</b> | <b>0.700</b> | 0.765        | 0.093     | 0.686        | 0.661        | 0.387        |
|                              | Hybrid         | <b>0.937</b> | 0.071        | 0.000        | 0.071        | 0.949        | 0.248     | 0.792        | 0.792        | 0.680        |
|                              | HAH            | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.848        | 0.025     | 0.697        | 0.697        | 0.633        |
|                              | Reasoning      | 0.396        | 0.056        | 0.040        | 0.056        | 0.336        | 0.032     | 0.232        | 0.150        | 0.084        |
|                              | Mini-Reasoning | 0.410        | 0.041        | 0.025        | 0.041        | 0.381        | 0.063     | 0.300        | 0.216        | 0.138        |
| Gemma-3-4B                   | Naive          | <b>0.937</b> | 0.049        | 0.000        | 0.049        | 0.799        | 0.125     | 0.694        | 0.664        | 0.387        |
|                              | Hybrid         | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.913        | 0.247     | 0.778        | 0.778        | 0.645        |
|                              | HAH            | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.614        | 0.039     | 0.560        | 0.560        | 0.462        |
|                              | Reasoning      | 0.394        | 0.049        | 0.031        | 0.049        | 0.331        | 0.029     | 0.306        | 0.177        | 0.096        |
|                              | Mini-Reasoning | 0.410        | 0.033        | 0.018        | 0.033        | 0.393        | 0.077     | 0.310        | 0.207        | 0.135        |
| Llama-3-8B                   | Naive          | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.700        | 0.110     | 0.646        | 0.607        | 0.358        |
|                              | Hybrid         | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.273        | 0.057     | 0.240        | 0.222        | 0.204        |
|                              | HAH            | <b>0.937</b> | 0.069        | 0.000        | 0.069        | 0.937        | 0.177     | 0.812        | 0.811        | 0.706        |
|                              | Reasoning      | 0.397        | 0.002        | 0.002        | 0.002        | 0.028        | 0.003     | 0.038        | 0.018        | 0.005        |
|                              | Mini-Reasoning | 0.410        | 0.011        | 0.006        | 0.011        | 0.093        | 0.010     | 0.092        | 0.059        | 0.028        |
| Phi-3-mini-4K                | Naive          | <b>0.937</b> | 0.005        | 0.000        | 0.005        | 0.862        | 0.303     | 0.705        | 0.805        | 0.679        |
|                              | Hybrid         | <b>0.937</b> | 0.095        | 0.000        | 0.095        | 0.924        | 0.267     | 0.780        | 0.780        | 0.675        |
|                              | HAH            | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.972        | 0.057     | <b>0.850</b> | <b>0.850</b> | 0.738        |
|                              | Reasoning      | 0.395        | 0.003        | 0.001        | 0.003        | 0.228        | 0.106     | 0.214        | 0.135        | 0.071        |
|                              | Mini-Reasoning | 0.410        | 0.003        | 0.001        | 0.003        | 0.233        | 0.102     | 0.196        | 0.118        | 0.062        |
| DeepSeek-R1-Distill-Llama-3B | Naive          | <b>0.937</b> | 0.094        | 0.032        | 0.094        | 0.890        | 0.425     | 0.750        | 0.750        | 0.524        |
|                              | Hybrid         | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.907        | 0.313     | 0.791        | 0.791        | 0.691        |

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| Model                        | Pipeline       | NDCG ↑       | R-1 ↑        | R-2 ↑        | R-L ↑        | Fluency ↑    | Coherence ↑  | Relevance ↑  | Factuality ↑ | Correctness ↑ |
|------------------------------|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|
|                              | HAH            | <b>0.937</b> | 0.000        | 0.000        | 0.000        | 0.900        | <b>0.439</b> | 0.756        | 0.756        | 0.637         |
|                              | Reasoning      | 0.393        | 0.013        | 0.007        | 0.013        | 0.288        | 0.096        | 0.240        | 0.145        | 0.087         |
|                              | Mini-Reasoning | 0.410        | 0.013        | 0.006        | 0.013        | 0.248        | 0.098        | 0.191        | 0.131        | 0.085         |
| AmbigQA                      |                |              |              |              |              |              |              |              |              |               |
| Model                        | Pipeline       | NDCG         | R-1          | R-2          | R-L          | Fluency      | Coherence    | Relevance    | Factuality   | Correctness   |
| DeepSeek-R1-3B               | Naive          | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.878        | 0.285        | 0.723        | 0.654        | 0.452         |
|                              | Hybrid         | <b>0.977</b> | 0.032        | 0.000        | 0.032        | 0.893        | 0.304        | 0.717        | 0.717        | 0.655         |
|                              | HAH            | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.902        | 0.314        | 0.746        | 0.746        | 0.677         |
| DeepSeek-R1-Distill-Llama-8B | Reasoning      | 0.435        | 0.008        | 0.001        | 0.008        | 0.402        | 0.125        | 0.328        | 0.096        | 0.079         |
|                              | Mini-Reasoning | 0.451        | 0.005        | 0.001        | 0.005        | 0.415        | 0.138        | 0.352        | 0.056        | 0.048         |
|                              | Naive          | <b>0.977</b> | 0.091        | 0.000        | 0.091        | 0.855        | 0.338        | 0.737        | 0.661        | 0.457         |
| Gemma-2-9B                   | Hybrid         | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.956        | 0.352        | 0.741        | 0.741        | 0.753         |
|                              | HAH            | <b>0.977</b> | 0.000        | 0.000        | 0.000        | <b>0.979</b> | 0.340        | 0.731        | 0.731        | <b>0.777</b>  |
|                              | Reasoning      | 0.435        | 0.001        | 0.000        | 0.001        | 0.150        | 0.057        | 0.149        | 0.060        | 0.040         |
| Gemma-3-12B                  | Mini-Reasoning | 0.451        | 0.001        | 0.000        | 0.001        | 0.142        | 0.048        | 0.137        | 0.061        | 0.046         |
|                              | Naive          | <b>0.977</b> | 0.080        | 0.000        | 0.080        | 0.793        | 0.132        | 0.754        | 0.634        | 0.376         |
|                              | Hybrid         | <b>0.977</b> | 0.062        | 0.000        | 0.062        | 0.814        | 0.126        | 0.744        | 0.739        | 0.631         |
| Gemma-3-4B                   | HAH            | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.733        | 0.044        | 0.651        | 0.638        | 0.604         |
|                              | Reasoning      | 0.438        | 0.071        | 0.000        | 0.062        | 0.409        | 0.036        | 0.326        | 0.153        | 0.088         |
|                              | Mini-Reasoning | 0.451        | 0.031        | 0.000        | 0.023        | 0.431        | 0.060        | 0.370        | 0.174        | 0.107         |
| Llama-3-8B                   | Naive          | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.848        | 0.274        | 0.748        | 0.653        | 0.413         |
|                              | Hybrid         | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.857        | 0.304        | 0.736        | 0.736        | 0.652         |
|                              | HAH            | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.614        | 0.065        | 0.529        | 0.529        | 0.498         |
| Phi-3-mini-4K                | Reasoning      | 0.440        | 0.115        | <b>0.061</b> | 0.114        | 0.435        | 0.044        | 0.269        | 0.146        | 0.087         |
|                              | Mini-Reasoning | 0.451        | 0.037        | 0.012        | 0.035        | 0.437        | 0.067        | 0.328        | 0.147        | 0.096         |
|                              | Naive          | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.794        | 0.229        | 0.744        | 0.640        | 0.391         |
| DeepSeek-R1-Distill-Llama-3B | Hybrid         | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.828        | 0.291        | 0.742        | 0.742        | 0.647         |
|                              | HAH            | <b>0.977</b> | <b>0.167</b> | 0.000        | <b>0.167</b> | 0.760        | 0.057        | 0.663        | 0.663        | 0.609         |
|                              | Reasoning      | 0.439        | 0.007        | 0.001        | 0.006        | 0.323        | 0.031        | 0.348        | 0.133        | 0.073         |
| Llama-3-8B                   | Mini-Reasoning | 0.451        | 0.001        | 0.000        | 0.001        | 0.407        | 0.074        | 0.374        | 0.116        | 0.072         |
|                              | Naive          | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.840        | 0.284        | 0.735        | 0.635        | 0.408         |
|                              | Hybrid         | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.554        | 0.198        | 0.489        | 0.464        | 0.435         |
| Phi-3-mini-4K                | HAH            | <b>0.977</b> | 0.050        | 0.000        | 0.050        | 0.879        | 0.151        | 0.785        | 0.784        | 0.687         |
|                              | Reasoning      | 0.438        | 0.001        | 0.000        | 0.001        | 0.147        | 0.040        | 0.164        | 0.048        | 0.037         |
|                              | Mini-Reasoning | 0.451        | 0.001        | 0.000        | 0.001        | 0.242        | 0.057        | 0.237        | 0.109        | 0.068         |
| DeepSeek-R1-Distill-Llama-3B | Naive          | <b>0.977</b> | 0.005        | 0.000        | 0.005        | 0.928        | 0.325        | 0.629        | 0.715        | 0.599         |
|                              | Hybrid         | <b>0.977</b> | 0.058        | 0.000        | 0.058        | 0.856        | 0.326        | 0.706        | 0.706        | 0.644         |
|                              | HAH            | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.931        | 0.076        | <b>0.811</b> | <b>0.811</b> | 0.741         |
| Llama-3-8B                   | Reasoning      | 0.436        | 0.001        | 0.000        | 0.001        | 0.244        | 0.112        | 0.261        | 0.065        | 0.035         |
|                              | Mini-Reasoning | 0.451        | 0.000        | 0.000        | 0.000        | 0.214        | 0.113        | 0.259        | 0.048        | 0.026         |
|                              | Naive          | <b>0.977</b> | 0.025        | 0.000        | 0.025        | 0.888        | <b>0.366</b> | 0.726        | 0.659        | 0.454         |
| Phi-3-mini-4K                | Hybrid         | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.865        | 0.355        | 0.719        | 0.719        | 0.643         |
|                              | HAH            | <b>0.977</b> | 0.000        | 0.000        | 0.000        | 0.823        | 0.332        | 0.724        | 0.724        | 0.631         |
|                              | Reasoning      | 0.437        | 0.006        | 0.000        | 0.006        | 0.330        | 0.106        | 0.285        | 0.076        | 0.053         |
|                              | Mini-Reasoning | 0.451        | 0.007        | 0.000        | 0.007        | 0.324        | 0.108        | 0.267        | 0.067        | 0.053         |

## C Technical Implementation Details

This appendix provides detailed implementation specifications for the components described in the main text.

### C.1 Reasoning Pattern Specifications

The Bayesian reasoning classifier relies on curated pattern sets for each reasoning type. Patterns are matched using precompiled regular expressions with word boundary constraints to prevent spurious substring matches.

**Factual patterns:** “what,” “who,” “where,” “when,” “which,” “list,” “define,” “name,” “identify,” “specify”

**Analytical patterns:** “how,” “analyze,” “examine,” “evaluate,” “explain,” “describe,” “assess,” “investigate”

**Comparative patterns:** “compare,” “contrast,” “versus,” “differ,” “similar,” “different,” “better,” “worse”

**Causal patterns:** “why,” “because,” “cause,” “effect,” “result,” “consequence,” “lead to,” “due to”

**Hypothetical patterns:** “if,” “would,” “could,” “might,” “assume,” “suppose,” “imagine,” “hypothetically”

The feature weights for the likelihood computation are:

| Type         | Entropy | Coherence | Coverage |
|--------------|---------|-----------|----------|
| Factual      | 0.4     | 0.4       | 0.2      |
| Analytical   | 0.3     | 0.4       | 0.3      |
| Comparative  | 0.2     | 0.4       | 0.4      |
| Causal       | 0.3     | 0.3       | 0.4      |
| Hypothetical | 0.2     | 0.5       | 0.3      |

Type-specific scaling weights reflecting prevalence and detection reliability, Factual = 1.4, Analytical = 1.25, Comparative = 1.0, Causal = 1.2, Hypothetical = 0.8.

## C.2 Score Fusion Methods

HAH-RAG implements multiple score fusion strategies beyond the default RRF:

**Weighted Linear:**  $S(d) = w_s \cdot \tilde{S}_{\text{sparse}}(d) + w_d \cdot \tilde{S}_{\text{dense}}(d)$  where  $\tilde{S}$  denotes min-max normalized scores and default weights are  $w_s = 0.4$ ,  $w_d = 0.6$ .

**Query-Adaptive:** Adjusts weights based on query characteristics. Queries with proper nouns or technical terms receive increased sparse weight (+0.2); long queries or those with WH-words receive increased dense weight (−0.2 from sparse).

**Score-Adaptive:** Uses retrieval score statistics to determine weights:

$$w_s = 0.6 \cdot \frac{\text{Var}(S_{\text{sparse}})}{\text{Var}(S_{\text{sparse}}) + \text{Var}(S_{\text{dense}})} + 0.4 \cdot \frac{\mu_{\text{sparse}}}{\mu_{\text{sparse}} + \mu_{\text{dense}}} \quad (34)$$

**Harmonic Mean:**  $S(d) = \frac{2 \cdot \tilde{S}_s(d) \cdot \tilde{S}_d(d)}{\tilde{S}_s(d) + \tilde{S}_d(d)}$

**Geometric Mean:**  $S(d) = \sqrt{\tilde{S}_s(d) \cdot \tilde{S}_d(d)}$

**CombMNZ:**  $S(d) = (\tilde{S}_s(d) + \tilde{S}_d(d)) \cdot |\{r : S_r(d) > 0\}|$

The automatic fusion method selection follows these rules:

- Query-Adaptive: proper noun ratio > 0.5 or technical term ratio > 0.3
- Harmonic Mean: word count > 15 and contains WH-words
- CombMNZ: query contains Boolean operators
- Score-Adaptive: default for all other queries

## C.3 Reranking Implementation

The cross-encoder reranker uses cross-encoder/ms-marco-MiniLM-L-12-v2 with the following configuration:

- Maximum sequence length: 512 tokens
- Default batch size: 64 (GPU) or 16 (CPU)
- Relevance threshold: 0.5
- Fallback: return unfiltered ranking when filtering produces empty set

Memory-efficient batch processing, When GPU memory is insufficient for the default batch size, the system reduces batch size by a factor of 4 and retries. If GPU processing fails entirely, the system falls back to CPU processing.

## C.4 Asynchronous Processing

HAH-RAG uses Python’s `asyncio` with `ThreadPoolExecutor` for concurrent operations:

- Sparse and dense retrieval execute in parallel via `asyncio.gather`

- Second-pass chunk searches execute in parallel with result merging
- Reranking batches can be processed concurrently when resources permit
- Deduplication preserves original ordering while eliminating redundant passages

### C.5 Generation Parameter Schedule

Token allocation based on context-to-capacity ratio  $\rho = (|C| + |q|)/L_{\text{model}}$ :

| Context Ratio            | Max New Tokens |
|--------------------------|----------------|
| $\rho < 0.40$            | 8192           |
| $0.40 \leq \rho < 0.55$  | 7168           |
| $0.55 \leq \rho < 0.625$ | 6144           |
| $0.625 \leq \rho < 0.70$ | 5120           |
| $0.70 \leq \rho < 0.78$  | 4096           |
| $0.78 \leq \rho < 0.86$  | 3072           |
| $0.86 \leq \rho < 0.94$  | 2560           |
| $\rho \geq 0.94$         | 2048           |

Frequency penalty adaptation based on input token count:

- $\leq 512$  tokens: penalty = 0.01
- 513 – 1024 tokens: penalty = 0.05
- $> 1024$  tokens: penalty = 0.10

### C.6 System Prompt Templates

Each reasoning type uses a specialized system prompt. The general template structure is:

[INST] <role> <analysis\_steps> <context> <question> [/INST]

**Factual prompt analysis steps:** (1) Identify key facts from context, (2) Extract relevant information, (3) Verify factual consistency, (4) Present information clearly and concisely.

**Analytical prompt analysis steps:** (1) Identify the core goal and constraints, (2) Break down key requirements, (3) Research factual information, (4) Verify accuracy, (5) Examine relationships, (6) Synthesize insights, (7) Draw evidence-based conclusions.

**Comparative prompt analysis steps:** (1) Identify elements for comparison, (2) Examine similarities and differences, (3) Evaluate relative strengths/weaknesses, (4) Draw balanced conclusions.

**Causal prompt analysis steps:** (1) Identify cause-effect relationships, (2) Examine contributing factors, (3) Analyze implications and consequences, (4) Establish causal chains.

**Hypothetical prompt analysis steps:** (1) Consider given conditions, (2) Analyze potential scenarios, (3) Evaluate implications, (4) Draw reasoned conclusions.

**Trivial prompt:** Minimal instructions for conversational exchanges, responding naturally without extensive analysis.